Supplement: Complete Proofs

From a Single Occurrence Source to a Continuous–Quantum Unified Field: Ten-Stage Closure and Universal Formation in a Spin×SU(7) Theory

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This Supplement provides complete, self-contained written proofs and verification notes for §4.3 (Theorems 4.6–4.7), §6.3–§6.5 (Theorems 6.7–6.11), §7.1–§7.5 (Theorems 7.1–7.12), and §10.1–§10.2 (Theorem 10.1, Propositions 10.2–10.5) of the main text. Sections S1–S9 cover the first-edition quantum construction and public experimental consumers; S11–S15 cover the second-edition CAR created state, nonlinear half-axis differentiability, nine-constraint Schur feedback, fourth-order holomorphic infrared return, emergent long-range potential, charged three-fermion positive spectrum, and 120 ns full-gate recovery; S16–S18 supply complete written proofs for the third-edition (v3) Volterra uniqueness and derivative-loss-free symmetric-hyperbolic all-order energy closure, 192/97 electromagnetic clock-factor cancellation and 4D curvature strong integration, and whole-word Kraus–Born isometry with open recycling density trace duality; S10 gives the three-edition claim and proof-location concordance. Proofs for §2–§4.2, §5, §6.1–§6.2, and §8–§9 appear in full in the main text and Appendices A–C (including Appendix B.8 for Theorem 3.3 and Appendix C.9 for Theorem 9.4).

The "Formal correspondence" note at the end of each section names the Lean 4 declarations or frozen receipts carrying the result; full commits, paths, and public reproduction entry points appear in Appendices B.7, C.8, D.7–D.9, and Code and data availability. Full SHA-1 hashes for fixed commit aliases **H**, **F**, **AB**, **AC**, **AD**, **AE**, **CAP**, **71e**, **9c73**, **df4**, **7e4**, **B291**, and **B300** are given in Appendices D.1, D.7, and D.9. Every numbered theorem is proved decisively within this paper and delegates nothing to unpublished companion manuscripts.

The fixed third-edition proof map records complete declaration types, direct consumers, and original receipts.

Verification credentials are classified into five explicit types: **formal theorems** (Lean 4 strict-kernel declarations), **finite exact computations** (finite identities decided inside the proof kernel), **fixed exact source programs with independent recomputation** (exact symbolic/algebraic programs paired at the same commit with zero-shared-code recomputations), **dual-implementation numerical verifications** (independent primary and secondary implementations producing frozen receipts with explicit error bounds), and **statistical contracts** (anytime-valid tests under pre-registered α-budgets).

S0 Common Notation and Tools

S0.1 Configuration and test spaces. Fix S_0 and the common action (3.1) on the connected component $\mathcal{U}_{103} = \mathcal{U}_q \times \mathcal{U}_x \times \mathbb{R}^{36}$ where $v_e > 0$, $\det h \neq 0$, $g_{00} \neq 0$, $h^{00} \neq 0$, $\det D_9 \neq 0$, and the 4×4 temporal–electric block $\det \mathcal{M} \neq 0$; write $\mathcal{U}_{100}$ for the three-stabilizer section (S1). Let $\mathcal{I} = \{+,-\} \times \{0,1,2,3\} \times (\Lambda^6 \sqcup \Lambda^2 \sqcup \Lambda^4)$ ($|\mathcal{I}| = 504$) and $\mathcal{F}_{504} = \Lambda^* \mathbb{C}^{504}$. The configuration Hilbert space $\mathcal{H}$ is given by the weighted L^2 direct sum (S1.5), in which $\mathcal{Q} = C_c^\infty(\mathcal{U}_{100}; \mathcal{F}_{504})$ is dense.

S0.2 Truncation compressions and retained terms. For a finite frame index F of the Gauss unitary history, $C_F(p)$ denotes the self-adjoint compression of the native Hamiltonian H_0 at momentum p, A_F the uncut Yukawa operator on the finite retained set, and $K_F(p) = C_F(p) + A_F$ the truncated generator (C_F preserves Λ^6 grading; A_F raises it by 1). Write $T_p(t) = e^{-itK_F(p)}$ and $R_p(z) = (K_F(p) - z)^{-1}$ ($\operatorname{Im} z \neq 0$). Truncation indices are directed by inclusion; a property holds "for cofinal F" if it holds for all F beyond some index along the source filter.

S0.3 Response points and the five-factor kernel. A **response point** q carries $(F, p_L, p_R, z, w, \varepsilon, T)$ ($\operatorname{Im} z \neq 0, \operatorname{Im} w \neq 0, \varepsilon > 0$) and a source field curve. With quantized first-variation current port J, the five-factor kernel is

$$\mathcal{K}_q(t) = T_{p_R}(-t)\, R_{p_R}(z)\, J\, R_{p_L}(w)\, T_{p_L}(t). \tag{S0.1}$$

$\mathbb{C}^{289}$ is the complexified space of real variations across the nine fields, and $H_{289}(p)$ is the second-variation matrix at four-complex momentum p (S5).

S0.4 Resolvent and Duhamel identities. Self-adjoint A satisfies $\|(A-z)^{-1}\| \leq |\operatorname{Im} z|^{-1}$ and $R_p - R_k = R_p(C_F(k) - C_F(p))R_k$; operator-norm Duhamel gives $e^{-itB} - e^{-itA} = -i \int_0^t e^{-i(t-s)B}(B - A)e^{-isA} ds$; and for $\Phi_t(X) = T_{p+k}(-t) X T_p(t)$, $\int_0^\infty e^{-\lambda t} \Phi_t \, dt$ converges absolutely in Bochner norm for all $\operatorname{Re} \lambda > 0$ under polynomial time growth.

S0.5–S0.6 Kernel verification of finite exact identities and data-bearing statements. Finite identities such as the 3071-entry dictionary of H_{289} over $\mathbb{Q}(\sqrt{2}, \sqrt{15})$, the 576 polynomial matrix inverse entries, and 289-dimensional matrix identities are proved inside the Lean 4 kernel via canonical table equality (`decide +kernel`), algebraic ring tactics (`ring`, `norm_num`, `field_simp`, `linear_combination`), and `fin_cases`, with zero floating-point approximations. Dual-implementation numerical and statistical clauses cross-check frozen primary and independent receipts under pre-registered criteria (S9, S15).

S1 Theorem 6.7: Common Quantum Hamiltonian

Objects and domain. Fix S_0 and (3.1) on $\mathcal{U}_{103}$ (S0.1). Matter consists of the 252-dimensional spinor–exterior field ψ and independent dual χ; dimensions $61, 504, 103, 100, 392$ are fixed uniquely by the source representations.

(1) Canonical carrier (61 CCR pairs and 504-mode two-branch CAR). - *Four-sector Legendre map*: when $\det h \neq 0$ ($h = e_{\mathrm{sp}}^{\mathsf{T}} \eta e_{\mathrm{sp}}$), an explicit rank-six right inverse yields the coframe

six-velocity relations

$$\Pi_e = M_e\dot{e} + b_e, \qquad Z^{\mathsf{T}}(\Pi_e - b_e) = 0, \qquad \dot{e} = Q_e(\Pi_e - b_e) + Z\lambda, \tag{S1.1}$$

where $M_e = D_h^{\mathsf{T}} B_h D_h$, $B_h = (4v_e)^{-1}\operatorname{Hess}_h \det h$, $Q_e = R_h B_h^{-1} R_h^{\mathsf{T}}$, and the ten columns of Z encode four temporal and six internal Lorentz primary constraints. Scalar momentum is $\Pi_\phi = h^{00}(\dot{\phi} + \rho^{70}_{A_0}\phi) + b_\phi$ ($h^{00} \neq 0$); spatial gauge momentum inverse is $K_E^{-1} = -\frac{\sigma v_e}{g_{00}}(g^{-1})_{\mathrm{sp}}$ (twelve temporal momenta vanish); and matter conjugate momentum is $p = -i\chi E$ ($E = v_e(e^{-1})^0{}_a i\gamma^a \otimes I_{63}$, without tying $\chi = \psi^\dagger$). The common canonical 1-form is

$$\Theta = \Pi_e \cdot \delta e + \Pi_\phi \cdot \delta\phi + \sum_i \Pi_A^i \cdot \delta A_i + \operatorname{Re}(ip\,\delta\psi), \tag{S1.2}$$

with twelve-dimensional Gauss constraint $\mathcal{G} = \mathcal{G}_A + \Pi_\phi^{\mathsf{T}}\rho^{70}\phi + \operatorname{Re}(ip\,\rho^{252}\psi)$. - *61 scalar CCR pairs*: the gauge–scalar orbit matrix $O \in \mathbb{R}^{70\times 12}$ has rank 9 ($\det D_9 = 256$ at the source for $D_9 = O_b^{\mathsf{T}} O_b$); choosing an orthogonal basis $R \in \mathbb{R}^{70\times 61}$ of $\ker O^{\mathsf{T}}$ and setting $\phi = v + Rx$, $\Pi_\phi = R^\vee \pi + O_b \zeta$ ($R^\vee = R(R^{\mathsf{T}}R)^{-1}$) pulls the scalar part of Θ back via $O_b^{\mathsf{T}} R = 0$ to $\pi \cdot \delta x$, yielding 61 standard CCR pairs $[x_j, -i\partial_{x_k}] = i\delta_{jk}$. - *504-mode two-branch CAR*: real coordinates $(\operatorname{Re}\psi, \operatorname{Im}\psi)$ and conjugate momenta $(-\operatorname{Im} p, -\operatorname{Re} p)$ of $\operatorname{Re}(ip\,\delta\psi)$ form two real branches whose single complexification quantizes over $\mathcal{I}$ ($|\mathcal{I}| = 504$) into one standard CAR algebra, with gauge current $Q_a = \operatorname{diag}(i\rho_a^{252}, i\overline{\rho_a^{252}})$ and matter Hamiltonian $\operatorname{diag}(M, -\overline{M})$. - *103-dimensional energy core*: the $6 + 61 + 36 = 103$ bosonic coordinates sum the four sector energies on

$$\mathcal{D}_{103} = C_c^\infty(\mathcal{U}_{103}; \mathbb{C}) \otimes \mathcal{F}_{504}, \tag{S1.3}$$

preserving arbitrary finite ordered compositions (including 944 nonzero cross-principal entries and 61 Yukawa force directions).

(2) Three-stabilizer section, second-class measure cancellation, and common Hamiltonian.

> **Lemma S1.1 (Kernel stabilizer commutation).** The three stabilizer generators S_a spanning $\ker O$ ($[S_a, S_b] = -2\epsilon_{abc}S_c$) generate operators $\widehat{G_a}$ on $\mathcal{D}_{103}$ satisfying $[\widehat{G_a}, \widehat{H}] = 0$, whose joint kernel is nontrivial and invariant under all finite compositions of the four-sector energy.

Proof. From $V_a D_9 = -C_a^{\mathsf{T}} D_9 - D_9 C_a$ one obtains $V_a F = C_a F + F C_a^{\mathsf{T}}$ ($F = D_9^{-\mathsf{T}}$), cancelling normal-coefficient and current-matrix variations slot by slot inside ordered squared momenta; gauge and coframe invariance follow from Gram invariance and spinor–exterior commutativity, with $|(7, 259)\rangle$ times a bump function witnessing a nonzero kernel state. ■

At the source point, the three gauge pivots have local orbit determinant $\det M_* = -54\sqrt{2}/125 \neq 0$, giving via the inverse function theorem a 100-dimensional open section $\mathcal{U}_{100}$ with extension $\mathcal{E}$ and restriction $\mathcal{R}$ ($\mathcal{R}\mathcal{E} = I$, $\widehat{H}\mathcal{E} = \mathcal{E}H_{100}$). The orbit Jacobian supplies positive weight $\rho_3(z) = 8a_1^2 a_{12} > 0$ ($54\sqrt{2}/125$ at the source), while the base volume, normal Gaussian momentum integral, and second-class determinant across the nine second-class scalar constraints cancel identically:

$$\frac{1}{256}\cdot\frac{256}{|\det D_9|}\cdot|\det D_9|=1. \tag{S1.4}$$

Assigning volume weight v^{m+2} to the m-particle sector defines

$$\langle f,g\rangle=\sum_{w\subset\mathcal{J}}\int_{\mathcal{U}_{100}}\overline{f_w(z)}\,g_w(z)\,\rho_3(z)v(z)^{m+2}\,dz,\qquad \mathcal{H}=\bigoplus_{w\subset\mathcal{J}}L^2(\mathcal{U}_{100},\rho_3 v^{|w|+2}dz), \tag{S1.5}$$

with surjective half-density isometry $(Uf)_w=\rho_3^{1/2}v^{1+m/2}f_w$ onto flat Lebesgue measure.

Lemma S1.2 (Formal adjoint pair and minimal closed graphs). On the dense invariant core $\mathcal{D}_{100}$, H_0 is formally symmetric, while $H=H_0+Y$ and $H^\sharp=H_0+Y^\dagger$ satisfy $\langle f,Hg\rangle=\langle H^\sharp f,g\rangle$ and are both closable with single-valued minimal graph closures; the $196\times 2=392$-mode two-sided Yukawa kernel reduces H_0 to a symmetric operator.

Proof. Integration by parts verifies symmetry of H_0 under v^{m+2}; only unidirectional Y conjugates to $Y^\dagger$ ($\|Y|144,396\rangle\|^2=4N^2=216/125\neq 0$). If $f_n\to 0$ and $Hf_n\to g$, then $\langle h,g\rangle=\lim_n\langle H^\sharp h,f_n\rangle=0$ for all $h\in\mathcal{D}_{100}$, forcing $g=0$ by density of $\mathcal{D}_{100}$. Half-density conjugation $UHU^{-1}=\mathrm{Op}_W(\sigma_0+Y)$ (merging coframe divergence and half-density potential into $N(3\mathrm{Number}^2+18\mathrm{Number}+20)/(16v)$) generates all 100 CCR and 504 normal-ordered CAR Heisenberg ports. ■

(3) Configuration coupling identification. Scalar Gram generates $4W(q)\geq 4$, and $K(p)$ satisfies slot by slot on $\mathbb{C}^{289}$

$$S_{\mathrm{phys}}(p)+Y=S_{\mathrm{nat}}+S_{\mathrm{cf}}+K(p)-K_{\mathrm{ret}}, \tag{S1.6}$$

which is Eq. (6.3) of the main text.

Formal correspondence. Sub-results of Theorem 6.7(1)(2) are fixed across four constrained-quantization commits (`ecd/` = `Verification/physics/low-energy-phenomenology/external-composite-decay/`, full hashes in Appendix D.7):

Sub-result	Commit	Producer and independent verification	Credential
61 CCR, 103-dim core, 504-mode CAR	`85cb5386...`	`ecd/source_{scalar_gauss_reduction,joint_local_quantum}.py` & `independent_*.py`	Exact source program + independent recomputation
3-stabilizer commutation (Lemma S1.1) & pivots	`f9b73392...`	`ecd/source_{quantum_stabilizer,stabilizer_phase_reduction}.py` & `independent_*.py`	Exact source program + independent recomputation

Sub-result	Commit	Producer and independent verification	Credential
Section, (S1.4), half-density $\mathcal{H}$, Lemma S1.2 & Weyl	`6726f385...`	`ecd/{SecondClassReduction,SourceQuantumConfigurationHilbert,SourceQuantumHalfDensityHilbert,FockFilteredWords,SymmetricGraphClosure,AntiunitaryDefectPair}.lean`; `ecd/source_{quantum_gauss_section,gauss_section_measure,full_gauss_section,yukawa_reducing_carrier,common_weyl_symbol}.py` & `independent_*.py`	Lean formal theorem + exact program & recomputation
100 CCR & 504 CAR Heisenberg ports	`eec031c4...`	`ecd/JointCCRCarPorts.lean`; `ecd/source_joint_ccr_car_ports.py` & `independent_*.py`	Lean formal theorem + exact program & recomputation

Theorem 6.7(3): `alpha-source/CanonicalPreparationActionCoreDecomposition.lean` (`physical_action_decomposition`) and `CanonicalPreparationFieldQuantization{Symbols,Core,Contacts}.lean` (commit AB).

S2 Theorem 6.8: All-Order Clock, Complete Fixed R, and Common Source Preparation

Objects and domain. $\mathcal{H}$, $\mathcal{D}_{100}$, $H = H_0 + Y$, and the invertible 4×4 temporal Jacobian J_4 on the source support cone.

(1) All-order clock recursion. The k-th order clock symbol over four temporal directions and $k+1$ slots starts from the source seed at $k = 0$ and recurses via

$$C_{k+1}(a, \text{last}) = -\sum_b (J_4^{-1})_{ab}\, R_{k+1,b}, \tag{S2.1}$$

where $R_{k+1,b}$ is built from 42 paid lower-order slot functions, ordered Moyal words, and 13 temporal leaves.

> **Lemma S2.1 (Force cancellation and lower-order preservation).** For every $k \geq 0$, C_{k+1} agrees identically with C_k on the first $k+1$ slots, annihilates all four $(k+1)$-st successor forces via $r + J_4(-J_4^{-1}r) = 0$ on the closed support, and preserves finite-order derivative budgets across all 100 coordinates. ■

(2) Complete fixed R. On Schwartz inputs, the full energy form splits into principal part plus integrable tail:

$$\mathcal{E}_B(g, f) = \mathcal{E}_{\text{prin}}(g, f) + \mathcal{E}_{\text{tail}}(B; g, f), \qquad R = E_{\text{tail}} - C_{\text{src}}. \tag{S2.2}$$

Lemma S2.2 (Bounded self-adjointness of R and Weyl readback). Row/column L^1 bounds S_1, S_2 of the shifted tail kernel give $\|E_{\text{tail}}\| \le \sqrt{S_1 S_2}$ by Schur's test, so R is bounded self-adjoint ($\|R\| \le \|E_{\text{tail}}\| + \|C_{\text{src}}\|$), satisfies

$$\langle Ag, Af\rangle + \langle g, Rf\rangle = \mathcal{E}_B(\check{g}, \check{f}), \tag{S2.3}$$

and keeps R, $\mathcal{T}$, and E strictly independent of ε. ■

(3) Common preparation and dual-residual vanishing. With $B = (\mathcal{T} + 1 + \|R\|)^{-1}$ and $\|B\| \in \sigma(B)$, set

$$E = \|B\|^{-1} - (1 + \|R\|). \tag{S2.4}$$

Lemma S2.3 (11-field common preparation point). For every $\varepsilon > 0$, pulling an approximate endpoint eigenvector of B back by B^{-1} yields x_ε satisfying $\|v_\varepsilon\| = 1$, $\|\mathcal{T}x_\varepsilon - Ex_\varepsilon\| < \varepsilon$, the weak-form identity, native Yukawa charge -1, eight external legs, two-time/frequency convergence, and geometric cutoff bound

$$\|Y_{\text{closed}} v_\varepsilon - \text{cut}_n(v_\varepsilon)\| \le (915/916)^{n+1} \cdot 916 \cdot b. \tag{S2.5}$$

Along $\varepsilon_n = 1/(n+1)$, both near-spectral and full-Yukawa residuals vanish simultaneously at fixed $(\mathcal{T}, R, E)$. ■

Formal correspondence. `alpha-source/CanonicalPreparationEngineProgram.lean` (`sourceEngine_generated`, `sourceEngine_preserves`), `CanonicalPreparationEngineCancellationForces.lean` (`sourceEngineForces_successor_zero`), `CanonicalPreparationFullEnergyForm.lean` (`actual_closed_form_complete_energy`), `CanonicalPreparationSourceRemainder.lean` (`sourceRemainder_selfAdjoint`, `sourceRemainder_readback`), `CanonicalPreparationSource{PreparedState,BoundarySequence,CausalState,NonlinearState}.lean` (`sourcePreparation_exists`, `preparationSequence_residual`, `preparationSequence_yukawa_residual`). Commit AB.

S3 Theorem 7.2: Varying-Field Response, Spectral Measure, and 0/1/2-Jet Cutoff Exchange

Objects and domain. Carrier of Theorems 6.7–6.8, fixed R, x_ε, and smooth field curves parametrized by source coordinate slices.

Lemma S3.1 (Finite retained set and pointwise stabilization of truncated jets). At each fixed curve, momentum p, finite frame F, nonreal z, and parameter r, the retained term acts solely on a finite index set depending on (F, p); hence for $n \ge N$, $Y_n = Y_\infty$ identically as finite-dimensional operators, and 0-, 1-, and 2-jets of resolvents, currents, contacts, insertions, and inverse contacts converge in norm to their uncut counterparts. ■

Lemma S3.2 (Exchange of prepared readouts with cutoff removal). Joint inner-product continuity across completed external legs and truncated resolvents

transfers Lemma S3.1 to values, first derivatives, and second derivatives of prepared responses and all 36 curvature readouts. ■

Decisive derivation. (1) The 106-dimensional inverse transports Jv^{m+2} and first/second half-density derivatives while preserving $\sum_g P_g P_F H_{\mathrm{phys}} P_F P_g$, generating the 289-component field response, $-RJR$, and second-order contact jets while cancelling moving-fibre density derivatives identically. (2) For $\gamma > 0$, Laplace responses inside radius r_* obey the quadratic remainder bound

$$\left\|\Phi(r) - \Phi(0) - r\Phi'(0)\right\| \le L^2\, s(f, g, \psi, p, n, \gamma)\, m(\gamma)\, |r|^2\, \|u\|^2. \tag{S3.1}$$

(3) Follows from Lemmas S3.1–S3.2. (4) For H_0 and core g, channel squared norms sum to $\|g\|^2$ at every finite filter stage, yielding a unique positive spectral measure μ_g; for bounded A and $\mu > 0$, retarded amplitudes lie in $L^1(\mathbb{R})$ in frequency and converge uniformly in real time.

Formal correspondence. AB: `alpha-source/CanonicalPreparationSourceNonlinearState.lean`, `CanonicalPreparationYukawaObservedLimit.lean` (`observed_{first,second}_exchange`, `curvature_{first,second}_exchange`), `CanonicalPreparationHalfDensityCompressionFeed.lean`; `external-composite-decay/Source{HamiltonianSpectralMeasure,BoundedInsertionTime}.lean`. AC: `CanonicalPreparation{SpatialFullFormFeed,JointMixedResponse,RawJointFiveTerms,OriginalPreparedGreen,PhysicalFeedbackField}.lean`.

S4 Theorem 6.10: Complete Charge–Current Ward Identity and Common Preparation

Lemma S4.1 (Pair-fibre normal-ordering expansion on CAR). For complex matrices A, B of any mode dimension, $Q(A) = \sum_{i,j} A_{ij} c_i^\dagger c_j$ satisfies

$$Q(A)Q(B) = Q(AB) + \Pi(A, B), \qquad \Pi(A, B) = \sum_{i,j,k,l} A_{ij} B_{kl}\, c_i^\dagger c_k^\dagger c_l c_j, \tag{S4.1}$$

where $\Pi(A, B)$ commutes with all occupation weights and satisfies $\Pi(A, B)\big|_{N=1} = 0$ on the one-particle sector.

Proof. Expand $c_j c_k^\dagger = \delta_{jk} - c_k^\dagger c_j$ and $c_j c_l = -c_l c_j$. ■

Decisive derivation. (1) Applying Lemma S4.1 fibrewise to $P_k(z)$ and Q_a yields Eq. (6.5) $P(k)Q_a = J_{\mathrm{sp}}(k, a) + J_{\mathrm{pair}}(k, a)$ with $J_{\mathrm{pair}} = 0$ on Λ^1. (2) Unprojected $[Q_a, H_{\mathrm{full}}]$ retains all four channels:

$$[Q_a, H_{\mathrm{full}}] = J_{\mathrm{cfg}}(a) + J_{\mathrm{sp}}(a) + J_{\mathrm{pair}}(a) + J_Y(a), \tag{S4.2}$$

with compression defects vanishing on cofinal F and two-resolvent responses read on x_ε obeying

$$\|\mathrm{resp}\| \le \|x\|(c_a b_n(w) + b_n(z) c_a + |z - w| v_a(n, z, w))\|y\|. \tag{S4.3}$$

(3) Twelve temporal directions generate bounded Q, cancelling $-rQ$ drifts between bare generator and retained kernel, with $\sum_i P_i^\dagger(c_i f) = -Q$ on the 106-dimensional orbit graph.

Formal correspondence. `alpha-source/CanonicalPreparationElectric{PairCurrent,CoreWard,WeightedGraph,PreparedWard}.lean`, `CanonicalPreparationTemporalGlobalCharge.lean` (commit AB).

S5 Theorem 7.1: Mother-Action Second Variation Recovers `H_289`

Lemma S5.1 (Smoothness and four-sector quadratic decomposition). Generated smoothly by twelve support nodes at Construction 4.1, the Fréchet second variation splits into 1650 quadratic monomials across gravity (234), gauge (406), scalar (220), and Dirac–dual (790), with all 95 constants traced to S_0.

Lemma S5.2 (3300 two-leg Fourier contributions merge to the 3071-entry dictionary). Collecting like terms among the 3300 signed Fourier legs over $\mathbb{Q}(\sqrt{2},\sqrt{15})$ matches the 3071-entry dictionary of $H_{289}(p)$ identically (`decide +kernel` and ring identities).

Lemmas S5.3–S5.4 (289-slot Euler pullback and five-block Legendre readback). Pulling back the nine residuals along all 289 unit directions equals value-minus-divergence slot by slot; verifying the two-sided inverse K of $\frac{1}{2}\omega^{\mathsf{T}}\mathcal{H}\omega$ across 576 polynomial products and substituting $\Omega = -KJ$ alongside coframe Legendre transformation and 8-direction Clifford CAR normal ordering cancels Number^2 cross-terms to recover kinetic, current, spin-squared, $-\frac{9\,\mathrm{lapse}}{8V}$Number, and volume-potential blocks. ■

Formal correspondence. `alpha-source/CanonicalSourcePropagation{OriginalHessianReturn,PreparedMotherEulerReturn}.lean` (`nativeActionFourierHessian_original`, `nativeJacobi_original`, `nativeHolonomicEuler_raw`). Commit AD.

S6 Theorem 7.4: Five-Factor Half-Axis Return, Imaginary Spectrum, and Causal Inverse

Objects and mother Green identity. With $\mathcal{P}_q(\lambda)X = \lambda X + \mathcal{A}_L X - X\mathcal{A}_R$ ($\mathcal{A}_L = -iK_F(p+k), \mathcal{A}_R = -iK_F(p)$), on the nonempty 103-dimensional active regular domain (`decide +kernel` rational right inverse),

$$G = \Xi\,(D^{-1} + P_a K_{\mathrm{ext}}^{-1})\,\Xi(-p)^{\mathsf{T}}, \qquad H_{289}\,G = I - \Xi(-p)^{-\mathsf{T}} P_{\mathrm{null}} \Xi(-p)^{\mathsf{T}}. \tag{S6.1}$$

(1)–(2) Finite-window integration by parts and 57-term polynomial half-axis. Since A_F raises grading by 1 and vanishes at power 57, Duhamel truncates at degree 56:

$$\|T_p(t)\| \le \sum_{n=0}^{56} \big(|t|\cdot\|A_F\|\big)^n. \tag{S6.2}$$

For $\mathrm{Re}\,\lambda > 0$ ($\eta = \mathrm{Re}\,\lambda/8$), $\Xi(-\lambda)^{\mathsf{T}}\Phi^+ = \Sigma^+ + B_0$ with finite-window error

$$\big\|\Phi^+ - \Phi_T^+\big\| \le \frac{2}{\mathrm{Re}\,\lambda}\, c\, e^{-(\mathrm{Re}\,\lambda)T/2}. \tag{S6.3}$$

(3) Two-half-plane inverses and imaginary spectrum. Right- and left-half-plane Laplace integrals invert $\mathcal{P}_q(\lambda)$ on both sides for $\operatorname{Re}\lambda \neq 0$, confining $\sigma(\mathcal{P}_q) \subseteq i\mathbb{R}$, with $0 \in \sigma(\mathcal{P}_q)$ at zero transfer.

(4)–(5) Nonautonomous Picard evolution and the `3/16` causal anchor. Variation of constants on $[0, T_*)$,

$$\partial_a U\big|_{a=0} = T_p(t) \int_0^t T_p(-s)\, \partial_a \mathcal{G}_a\big|_0(s)\, T_p(s)\, ds, \tag{S6.4}$$

splits at anchor $\lambda_0 = 3 - g$ ($g = 1/[8(1+B)]$) into Laplace current ($\leq 1/16$) plus initial block ($\leq 1/8$), giving

$$\|\mathcal{U}\| \leq \tfrac{1}{16} + \tfrac{1}{8} = \tfrac{3}{16} < 1, \qquad \|(I - \mathcal{U})^{-1}\| \leq \tfrac{16}{13}, \tag{S6.5}$$

hence $\det(I - \mathcal{U}) \neq 0$ and a unique fixed point.

Formal correspondence. AC: `CanonicalPreparation{OriginalPreparedGreen,PhysicalFeedbackField}.lean`; AD: `CanonicalPhysicalLaplace.lean`, `CanonicalPreparationPhysical{FullHalfAxis,ControlledFieldTail}.lean`, `CanonicalSourcePropagation{AxisReadback,SourceSpectralInverse,ActualNativeHistoryReturn}.lean`; AE: `CanonicalPreparationSource{HalfAxisCurrentOperator,NativeFourierHistory,CurrentFeedbackAdjugate,AnalyticRegularPoint}.lean`.

S7 Theorem 7.7: Common Moving Carrier, Momentum Continuity, and Native Static Residue

1. **Moving eight-state overlap unitarity**: $U(p)^\dagger U(p) = I_8$ ($U(0) = I_8$) and $M(p_L, p_R)[A] = U(p_L)^\dagger M(0,0)[A] U(p_R)$.
2. **Affine momentum continuity**: $C_F(p) = C_F(0) + v_F(p)$ gives $\|R_p(z) - R_k(z)\| \leq |\operatorname{Im} z|^{-2} \|v_F\| \|p - k\|$ with p-invariant A_F.
3. **Punctured static pole and 114th colour cosource**: along $p_\kappa = (0, i\kappa, 0, 0)$, leading $-\kappa^2 S$ splits the uncleared Green function on the nonempty punctured domain into

$$G(\kappa) f = \text{contact} + (-\kappa^2)^{-1} V_0 S^{-1} V_0^{\mathsf{T}} f + \text{98-dim regular complement}, \tag{S7.1}$$

with $-\kappa^2 \varphi(\kappa) \to E_0(-\frac{9}{125}\sqrt{30}\mathfrak{w}, -\frac{67}{72}\sqrt{30}\mathfrak{w}, 0, 0, 0)$, where

$$\mathfrak{w} = \tfrac{3\sqrt{2}}{10}(J_{21} - J_{34}) = \kappa_{\mathrm{ct}} - \delta_{\mathrm{cf}} = -c_{114}(0, T), \qquad |\delta_{\mathrm{cf}}| \leq \frac{\|D_0 F\|\, |T|}{|\operatorname{Im} z|\, |\operatorname{Im} w|}, \tag{S7.2}$$

and $H_{289}\varphi = w_{\mathrm{win}} \iff$ all nine null cosources vanish ($R_b L = I_9$).

Formal correspondence. AE, CAP: `alpha-source/CanonicalPreparationSource{StaticPoleDomain,StaticOriginalGreen,StaticActualPole,ModeCurrent,ConstraintRead,ColorBoundaryWard,N1NormalWard,MovingCarrier,MomentumContinuation,FullPoleTensor,ModeContactRead,NativePoleBalance,NativePolePrice,AxisCompatibility}.lean` and five `AuditCanonicalPreparation*.lean` modules.

S8 Theorem 7.10: Matched Whole-Word Budget and Renormalized Endpoint Common Tail

With $H_0 = C_F + d_F$, $\sigma = \mu > 0$, $n = \sqrt{54/125}$, and $C_3 = UD + 3iD_cU$:

1. **Matched positive source and phase absorption**: $D_cU - UD_c = 2iU$ yields $J_3 = R_3 + 6\sigma\,\mathrm{Im}\langle C_3 w, w\rangle$ and the five-slot lower bound ($P_3(w) \le 4J_3$). Cauchy–Schwarz at $\eta = 1/2$ gives $J_3 - 2R_3 \le (96\sigma^2/n)\|w\|^2$, hence $\int_{\mathbb{R}}(J_3 - 2R_3)_+ dt \le \varepsilon$ beyond N_0, closing the $4\times6\times2 = 48$ joint budget via $\|v_\varphi\|^2 \le (1 + \frac{2}{25}\|\mathrm{vac}\|^2)\|w\|^2 + J_3/(20n)$.
2. **Renormalized second-Green $L^1(\mathbb{R})$ common tail and diffusion split**: reading fixed $H_0^2 g_i$ on cofinal F via $\int_{\mathbb{R}}(t^2+\mu^2)^{-1}dt = \pi/\mu$ ($\eta = \varepsilon\mu/(8\pi)$) and Ritt constants $(32, 256)$ bounds $\int_{\mathbb{R}} |\mathcal{E}|d\omega \le \varepsilon$ across both causal ports, while $\Gamma_L(X) = \mathcal{D}^{\mathsf{T}}X + X\mathcal{D} - 2AXA$ satisfies $\Gamma_L(1) = 0, \Gamma_L(V) = 18$ and $R_3 = \mathcal{E}/\sigma^2 - D_{\mathrm{ren}}/(2\sigma^2) + R_{\mathrm{df}}$.

Formal correspondence. AE (and `fa15016c...`): `external-composite-decay/SourceClockPhi{NativeMatchedSource,NativeMatchedGammaBudget,RenormalizedSecondGreenTail,MatchedDiffusionSource,FixedSourceJets,RittHigherGradient}.lean`.

S9 Public Experimental Credentials for Propositions 10.2–10.3, 10.5, and §10.2

S9.1 Proposition 10.5: Full Optical Source Domain and Strictly Positive Joint-95% CH

1. **Optical source domain and all-order Born closed form (formal theorem)**: under two-mode squeezed polarization parameters (m, z, x, r, e), sign invariants $k = (1-2\lambda)T$, quadratic bounds, and six training intervals hold iff lawful $\lambda = (1 - k/T)/2$ exists; four marginals invert linearly to covariances, and summing $\Gamma_n = O^\dagger X^{\otimes n} O$ geometrically over occupation pairs gives

$$Q_{AB} = \frac{1}{1 + n_V\,(T_A + T_B - T_A T_B)}. \tag{S9.1}$$

2. **Cover trees, held-out counterexamples, and strictly positive joint-95% CH (dual verification + statistical contract)**: primary and independent trees evaluate $81,922$ nodes each (finding 141 training-compatible held-out-violating members and 64 full-12-interval members: 41 positive and 23 negative CH). Under the 1/20 joint budget, `NormalizedSourceBet.lean` certifies $1 + 2^{-k}h > 0$ for

$$h = \ell \cdot \mathrm{CH} - \varepsilon(1-\varepsilon)\min(c_{01}^A, c_{10}^B, c_{11}), \qquad \log E_k = W_{\mathrm{obs}} \log a_k + L_{\mathrm{obs}} \log b_k - T \log(1 + 2^{-k}h), \tag{S9.2}$$

cross-checking 576 intervals and $23,040$ directional bets to confirm a nonempty joint 95% domain where every accepted source satisfies

$$\mathrm{CH}_{N5} > 1.3790180504 \times 10^{-6}. \tag{S9.3}$$

S9.2 Held-Out Slices, Finite Angular Gain, and Nominal Controls of §10.2

1. **Two-cell training slices and Bob's angle shift**: 69 seamless slices trained on 00 and 11 place held-out 01 and 10 envelopes inside the published intervals:

$$01 : [0.00015821397237,\ 0.00015855716089], \qquad 10 : [0.00015026944774,\ 0.00015053199993], \tag{S9.4}$$

while shifting Bob's angles by -0.01° on $\pm 0.05^\circ$ boxes yields strictly positive CH gain ($> 1.99 \times 10^{-9}$ and $> 1.69 \times 10^{-9}$; independent Fock endpoint gain 3.8493×10^{-9}).

2. **Historical nominal negative controls**: ETH zero-radius rejection appears in Appendix D.4; for NIST 2015 (0.003 efficiency half-width), the predeclared optimal band

$$r \in [0.30515678, 0.32571200],\ \ \theta_0 \in [4.727559^\circ, 5.179646^\circ],\ \ \theta_1 \in [-27.552834^\circ, -26.900901^\circ] \tag{S9.5}$$

excludes all five published nominal parameters (`REPLAY_DEVIATION_EXCEEDS_PREDECLARED_BAND`).

S9.3 Proposition 10.2: Zero-Input Tsirelson Lock, Readout Domain, and Equal-Law Quotient

1. **Cauchy–Schwarz Tsirelson bound**: writing $S_{\mathrm{CHSH}} = \langle u, v \rangle$ on $\mathbb{R}^4$ with $\|u\|^2 = 2, \|v\|^2 = 4$ gives $|S_{\mathrm{CHSH}}| \le \sqrt{8} = 2\sqrt{2}$, attained at $\pm 1/\sqrt{2}$.
2. **12-dim effect bijection, Munich 20,403 passage, and maximal quotient**: Bloch decomposition bijects onto the compact 12-dim domain; all 20,403 Munich prefixes stay below 40 under $\alpha = 1/20$ (whereas the ideal zero-error point crosses 40 at April trial 3974); rank-one factorization parameterizes equal-law regular settings by a unique nonzero inverse scale (s, t), with effective gain ≥ 0.36579.

S9.4 Proposition 10.3: Two-Probe Unique Inverse, Detector Fibre, and 12-State Pulse Image

1. **Two-probe inverse and detector fibre**: Cramer's rule inverts signed two-probe responses uniquely whenever the identification determinant is nonzero; positive-gain click effects $E_{\mathrm{click}} = dI + (1-d)\eta J$ are characterized by

$$0 \le d \le m, \qquad M - d \le k \le 1 - d \qquad (m = (1-\mu-g)/2,\ M = (1-\mu+g)/2), \tag{S9.6}$$

with unique inverse atomic effect and $\eta \lambda_{\max}(J) \ge 2g/(1+\mu+g)$.

2. **Pulse inequalities, Rabi-area ceiling, and 12-state verification**: rectangular pulse realizability ($\gamma = b - d > 0$) holds iff

$$(u^2 + z^2)(b+d)^2 \le (1-\mu)^2\gamma^2, \qquad (u^2+z^2)(2-b-d)^2 \le (1+\mu)^2\gamma^2; \tag{S9.7}$$

under $p_{\max}^{\mathrm{at}} \le 7\mathcal{A}^2/4$, gain obeys

$$g \le \min\Big(1,\ \frac{(1+B)\,\eta_{\mathrm{up}}\,\bar{a}}{2 - \eta_{\mathrm{up}}\,\bar{a}}\Big), \qquad \bar{a} = \min\Big(1, \frac{7A_{\mathrm{up}}}{4}\Big), \tag{S9.8}$$

verified across 32 role domains on Garthoff's 12-state master equation alongside six equal-law / distinct-spectrum pairs and 41,673 parsed local records.

Formal correspondence (S9.1–S9.4). Commits AB, AC, AD, AE: full paths in Appendix D.8.

S10 Three-Edition Claim and Proof-Location Concordance

To facilitate comparison across Zenodo v1 (`phys.P1-P36`), v2 (`phys.P37-P40`), and the present v3 overhaul (eliminating `a/b` patch numbers and adding seven self-contained theorems), the table below maps every claim group to its v3 location and complete proof:

Claim group	v1 / v2 IDs	Third-edition (v3) main text	Complete written proof	Commits
`phys.P1-P2, P19` Nine-field solution, uniqueness & bounds	v1: 4.1–4.5	Construction 4.1, Thms. 4.2–4.5 (§4.1–4.2)	§4.1–4.2; App. B.1–B.7	H
New in v3 Bidirectional spectral selection of $\mathfrak{h}_{\mathrm{SM}}$	(Block ansatz only)	**Theorem 2.2** (§2.2)	§2.2; App. A.1, B.7	H
`phys.P3, P13-P14` Exterior matter, bundle, vacuum, B_A	v1: 2.2–2.5, 3.2–3.3	Props. 2.3–2.6, 3.5; Thm. 3.4	§2.3–2.4, §3.3; App. A.3, B.2–B.6	H
New in v3 Nonseparability, four-leg EC & `5/8` defect	(Not in main text)	**Prop. 3.2, Theorem 3.3** (§3.1–3.2)	§3.1–3.2; **Appendix B.8**	H
New in v3 Volterra uniqueness & all-order energy closure	(Not in main text)	**Theorems 4.6–4.7** (§4.3)	**Supplement S16.1–S16.4**	H
`phys.P15` Clock–B cubic coupling & recoil	v1: 5.1–5.4	Construction 5.1, Thms. 5.2–5.4 (§5)	§5.1–5.3	H
`phys.P4-P5, P16` $4\sqrt{2}$ pairing & 256-dim Fock	v1: 6.1–6.6	Def. 6.1, Thms. 6.2–6.5, Ex. 6.6 (§6.1–6.2)	§6.1–6.2; App. B.4, D.3	H
`phys.P26` Common H, clock R, jets & Ward	v1: 6.7–6.10	Thms. 6.7–6.8, 6.10, 7.2 (§6.3–6.5, §7.1)	S1–S4	Early 4, AB, AC
`phys.P37` Created state, 12-dir norm. & $1/(4\pi r)$	v2: 6.8a, 6.10a, 6.13b	**Theorems 6.9, 6.11, 7.9** (§6.4–6.5, §7.4)	S11.1–S11.6	71e, 9c73
`phys.P27, P29-P30, P33, P36` H_{289}, half-axis & residue	v1: 6.11–6.13	**Theorems 7.1, 7.4, 7.7** (§7.1–7.2, §7.4)	S5–S7	AC, AD, AE, CAP
`phys.P38` Nonlinear derivative, 9-Schur & $z^4\Pi_{\mathrm{st}}$	v2: 6.9a, 6.12a, 6.13a	**Theorems 7.3, 7.5, 7.6** (§7.2–7.3)	S12.1, S12.4, S13	71e, 9c73

Claim group	v1 / v2 IDs	Third-edition (v3) main text	Complete written proof	Commits
New in v3 192/97 fraction, Ward factor, cone & 4D curvature	(Not in main text)	**Theorems 7.8, 7.12** (§7.4–7.5)	**Supplement S17.1–S17.4**	df4, 7e4
`phys.P34, P39` Matched tail & 3-fermion positive spectrum	v1: 6.14; v2: 6.15	**Theorems 7.10, 7.11** (§7.5)	S8, S14.1, S14.3	AE, 71e, 9c73
`phys.P7-P8, P12` Orbits, Legendre, 5-point & Bell	v1: 7.1–7.4	**Theorems 8.1–8.3, Prop. 8.4** (§8)	§8.1–8.2	H
`phys.P6, P9-P11, P17-P18, P20` Family, 10-stage seal & fold	v1: 8.1–8.3, App. C	**Theorems 9.1–9.3** (upgraded with 5-pos. assembly, zero authority & fold), App. C	§9.1–9.2; App. C.1–C.8	H
`phys.P22` Universal fixed-mother formation & no magic bit	v1: 8.4–8.5	**Theorem 9.4, Cor. 9.5** (§9.3)	§9.3; App. C.9	F
New in v3 Whole-word Kraus–Born & open density duality	(Not in main text)	**Theorem 10.1** (§10.1)	**Supplement S18.1–S18.2**	B291, B300
`phys.P24-P25, P28, P31-P32, P35, P40` Public Bell & 120 ns	v1: 9.1–9.3; v2: 9.4	**Props. 10.2–10.5**, §10.2, App. D.4	S9.1–S9.4, S15, S15.5	AB, AC, AD, AE, 71e, 9c73

S11 Theorems 6.9, 6.11, and 7.9: Actual Created State, Twelve-Direction Temporal Normalization, and Emergent Long-Range Potential

S11.1 Nonzero CAR Creation Lower Bound and Unit External State

Let $c_{12}(z)$ denote coordinate $(1,2)$ of the actual scalar field: its value at the source vacuum is 1, and inside the closed source box perturbation norms are strictly $< 1/2$, giving $|c_{12}(z)| \geq 1/2$ by the triangle inequality. The unoccupied colour annihilation mode a_{02} ($\|a_{02}\| \leq 1$) anticommutes with the other two colour creators, annihilates the seed state, and has unit anticommutator on its own colour, yielding pointwise

$$a_{02}\,\mathcal{A}_{10}^{\dagger} f = \overline{c_{12}}\, f, \qquad \int \|f(z)\|^2 \rho(z)\, dz \;\leq\; 4 \int \|\mathcal{A}_{10}^{\dagger} f(z)\|^2 \rho(z)\, dz. \tag{S11.1}$$

Extending by density to the unit background v_ϵ of Theorem 6.8 gives $\|\mathcal{A}_{10}^{\dagger} v_\epsilon\| \geq 1/2$, hence $\|g_\epsilon\| \geq 1/4 > 0$ for $g_\epsilon = (i/2)\mathcal{A}_{10}^{\dagger} v_\epsilon$; dividing by $\|g_\epsilon\|$ produces the unit state $u_\epsilon = g_\epsilon/\|g_\epsilon\|$ with $\|\ell_d\| \leq \|u_\epsilon\|^2 + \|v_\epsilon\|^2 = 2$.

S11.2 Joint Standard Model Charge Characters, Twelve-Direction Normalization, and Weighted Creation Current

For any $a = (A_3, A_2, y_a) \in \mathfrak{h}_{\mathrm{SM}}$ ($A_3^* = -A_3, \overline{y_a} = -y_a$), the four-form creation coefficients and 504-mode CAR commutator of L_a satisfy

$$c_r(a\Phi) = -\sum_s (A_3)_{sr} c_s(\Phi), \qquad [Q(L_a), a^\dagger_{s,r}] = \sum_v (A_3)_{vr} a^\dagger_{s,v} + y_a a^\dagger_{s,r}. \tag{S11.2}$$

Substituting both into $\delta_a^{\mathrm{joint}} \mathcal{A}^\dagger = \mathcal{A}^\dagger(a\Phi) - [Q(L_a), \mathcal{A}^\dagger(\Phi)]$ and using $\overline{(A_3)_{rs}} = -(A_3)_{sr}$ cancels the colour sums identically, leaving $-y_a \mathcal{A}^\dagger$ (and $+y_a \mathcal{A}$ for annihilation) on both the dense core and completed profile.

Across all twelve temporal directions, noncharacteristicity guarantees the canonical phase inverse of $\mathsf{w}_t(s)$,

$$M_{\mathrm{ph}}(s)\,[\mathsf{a}\,\mathsf{w}_t(s)] = I_{504}, \tag{S11.3}$$

which right-multiplies by C_a and expands via Lemma S4.1 to prove Eqs. (6.6)–(6.7). For any 504×504 weight W, setting $A^\dagger_W = [Q(W), \mathcal{A}^\dagger(\Phi)]$ and $A^\dagger_{a,\mathrm{sc,W}} = [Q(W), \mathcal{A}^\dagger(a\Phi)]$ and substituting $[Q(M), a_j^\dagger] = \sum_i M_{ij} a_i^\dagger$ with (S11.2) yields

$$[Q(WC_a), \mathcal{A}^\dagger(\Phi)] = iA^\dagger_{a,\mathrm{sc,W}} + iy_a A^\dagger_W, \tag{S11.4}$$

which evaluates at $W = \mathsf{w}_{\mathrm{source}}(z)$ on seed q to give Eq. (6.8).

S11.3 Temporal Field-Port Matching and Two-Sided Truncated Resolvent Tails

The full temporal field port J_a and fixed-momentum Noether port N_a satisfy $J_a = -N_a + C_a^{\mathrm{comp}}$. Writing truncated/uncut left and right inverses as L_c, R_c, L_u, R_u with differences plus retained leaks D_L, D_R, expanding $L_c = L_u + L_c D_L L_u$ and $R_c = R_u + R_c D_R R_u$ yields

$$L_c J_a R_c = -L_u N_a R_u + L_c C_a^{\mathrm{comp}} R_c - L_c D_L L_u N_a R_c - L_u N_a R_c D_R R_u. \tag{S11.5}$$

Invariance of the u_ϵ, v_ϵ subspace under R_u kills the right leak, and finite-frame stabilization gives bilinear convergence across all 289 ports.

S11.4 Same-Density Electromagnetic Pole Idempotence and 72 Gauge Curvatures

Writing $(\omega - \omega_*)G(\omega) \to R_\omega$ at the simple pole ω_*, multiplying $KG = I - Q_s$ and $GK = I - Q_f$ by $\omega - \omega_*$ gives $K_* R_\omega = R_\omega K_* = 0$; together with $R_\omega Q_s(\omega_*) = 0$, the difference quotient satisfies

$$R_\omega \frac{K(\omega) - K_*}{\omega - \omega_*} [(\omega - \omega_*)G(\omega)] = R_\omega [I - Q_s(\omega)] \;\longrightarrow\; R_\omega, \tag{S11.6}$$

proving $R_\omega K'(\omega_*) R_\omega = R_\omega$ (Eq. (7.15)). Classical window forcing of the form $K \cdot$ field is annihilated by $R_\omega K_* = 0$, leaving $R_\omega J_d$ read out continuously across all $6 \times 12 = 72$ non-Abelian curvature components

$$\mathcal{F}_{\mu\nu}(P;a) = P_\mu a_\nu - P_\nu a_\mu + [\bar{A}_\mu, a_\nu] + [a_\mu, \bar{A}_\nu]. \tag{S11.7}$$

S11.5 Emergent $1/(4\pi|x-y|)$ Spatial Kernel

Set $S_0 = -F_0 S_{\mathrm{st}}^{-1} F_0^{\mathsf{T}}$ from S7. Across all unit directions n, $r^2 G(0, irn) = r^2 A(0, irn) - F(0, irn) B(r, n)^{-1} F(0, -irn)^{\mathsf{T}}$ converges linearly to S_0. Joint continuity of uncut $L(k, M) = J_D^{\mathrm{uncut}}(-k)^{\mathsf{T}} M J_S^{\mathrm{uncut}}(k)$ generates $\rho_{DS} > 0$ and $B_{DS} = 1 + |L(0, S_0)|$ such that for $0 < a|k| < \rho_{DS}$,

$$\big|L(ak, a^2 G(0, iak))\big| \le B_{DS}|k|^{-2}, \qquad \lim_{a \downarrow 0} L(ak, a^2 G(0, iak)) = |k|^{-2} L(0, S_0). \tag{S11.8}$$

Local integrability of $|k|^{-2}$ on $\mathbb{R}^3$ and rapid decay of $\phi \in \mathcal{S}(\mathbb{R}^3)$ provide dominated convergence, and the Gaussian heat-kernel identity

$$\frac{1}{|k|^2 + d^2} = \int_0^\infty e^{-t(|k|^2 + d^2)} dt, \qquad \mathcal{F}^{-1}\big(e^{-t|2\pi\xi|^2}\big)(y) = (4\pi t)^{-3/2} e^{-|y|^2/(4t)} \tag{S11.9}$$

integrates at $d \downarrow 0$ to $1/(4\pi|y|)$, proving Eqs. (7.16)–(7.17).

S11.6 Born Overlap Dressing for Three-Fermion External States

On the three-fermion state q of S14.1 ($(-R_{0,F}Y)^4 b = 0$), orthogonal decomposition $r = \sum_{j=0}^{3} (-R_{0,F}Y)^j b$ defines $z = \|b\|/\|r\| \in (0, 1]$; with at least one formal-adjoint ($H^\sharp$, $z = 1$) external leg at both source and detector, grading-preserving gauge action equates full and base numerators, giving $C_{\mathrm{full}} = Z C_{\mathrm{base}}$ ($Z = \prod_{j=1}^{4} z_j$, base constant $9/125 + 67/72 = 9023/9000$).

Formal correspondence. Commits 71e, 9c73: `Verification/physics/low-energy-phenomenology/alpha-source/em-identification/` (`ActualDressed{SourcePreparation,TemporalNormalization,ReaderMatching,UncutMovingCoulomb}.lean`, `PhysicalDressed{P286Character,WeightedCreation}.lean`, `ActualEM{GaugeCurvature,DressedGaugePole}.lean`, `ActualCompositeCoulombDressing.lean`).

S12 Theorems 7.3 and 7.6: Constant-Field Nonlinear Half-Axis Derivative and Holomorphic Matrix Infrared Return

S12.1–S12.3 Identification of the Nonlinear Half-Axis with the Perturbed Pencil Inverse and C^2 Fréchet Derivative

For $h \in \mathbb{R}^{289}$, put $I_i(h) = R_+(h) N_i(h) R_-(h)$ and $\mathcal{P}_\lambda(h)X = \lambda X + A_+(h)X - X A_-(h)$ ($A_\pm = -iK_\pm(h)$). Bounded invertibility of $\mathcal{J}_\lambda = \mathcal{P}_\lambda(0)^{-1}$ for $\mathrm{Re}\,\lambda > 0$, C^2 continuity, and Duhamel–Gronwall ($\eta = \mathrm{Re}\,\lambda/4$) generate an open ball around 0 where $\|\mathcal{J}_\lambda[\mathcal{P}_\lambda(h) - \mathcal{P}_\lambda(0)]\| < 1/2$ and

$$\|e^{-\lambda t}\mathcal{T}_h(t)\| \;\le\; \mathcal{B}_d(\mathrm{Re}\,\lambda/4)\, e^{-(\mathrm{Re}\,\lambda)t/2} \qquad (\forall t \ge 0). \tag{S12.1}$$

Integrating by parts (boundary vanishing by (S12.1)) identifies the nonlinear integral with the perturbed operator inverse:

$$\int_0^\infty e^{-\lambda t}\mathcal{T}_h(t)\,dt = \mathcal{P}_\lambda(h)^{-1}, \qquad \mathcal{J}_d(h;\lambda)_i = \ell_d\{\mathcal{P}_\lambda(h)^{-1}I_i(h)\}. \tag{S12.2}$$

Since operator inversion is C^2 on the invertible open set, differentiating (S12.2) at $h=0$ along any $f\in\mathbb{R}^{289}$ via $D_h\mathcal{P}_\lambda^{-1}(0)[f] = \mathcal{I}_\lambda\mathcal{D}_f\mathcal{I}_\lambda$ yields

$$D_h\mathcal{J}_d(0;\lambda)[f]_i = \ell_d\,\mathcal{I}_\lambda\{I^N_{i,f} - J_L[f]\mathcal{I}_\lambda I_i(0) + [\mathcal{I}_\lambda I_i(0)]J_R[f]\}, \tag{S12.3}$$

matching $\Pi^{\mathrm{st}}_d f$ column by column (Eq. (7.3)), while tail integration gives (7.4) and nonautonomous five-term differentiation on $[0,\tau)$ gives (7.5).

S12.4–S12.5 Occupation-Graded Cubic Time Bound and All-Branch $z^4\Pi_{\mathrm{st}}$ Holomorphic Return

Since $C_F(h)$ preserves grading and $Y_F(h)$ raises it by 1, two-particle u_ϵ climbs at most twice and one-particle v_ϵ at most once, giving the three-stage Duhamel cubic bound

$$\|\partial_f T_F(t)v\| \;\le\; |t|j(1+2a|t|+3a^2|t|^2)\|v\| \;\le\; 3j(1+a)^2(1+|t|)^3\|v\|. \tag{S12.4}$$

Moreover, the zero-climb projector P ($Pu_\epsilon = u_\epsilon, Pv_\epsilon = v_\epsilon$, $[P,C_F(h)]=0$, $PY_F(h)=0$) satisfies $PT_{\mathrm{full}}(h,t) = PT_C(h,t)$ and $PR_{\mathrm{full}}(h,z) = PR_C(h,z)$, reducing the left time and resolvent legs of the five-factor kernel to norm-bounded free propagators and proving $|\mathcal{D}_{ij}(t)| \le K_{ij}(d)(1+|t|)^3$. Integrating $\int_0^\infty e^{-\sigma t}(1+t)^3dt = \sigma^{-1}+3\sigma^{-2}+6\sigma^{-3}+6\sigma^{-4}$ on $0<\sigma\le 1$ gives $\sigma^4|\Pi^{\mathrm{st}}_d(0,\sigma)_{ij}| \le 56K_{ij}(d)$ (Eq. (7.9)).

At zero transfer, substituting $\mathcal{I}_z = z^{-m_d}T_d(z)$ from the minimal polynomial $p_d(z) = z^{m_d}d_d(z)$ ($d_d(0)\neq 0$) into (S12.3) defines holomorphic

$$R_d(z) = z^{2m_d}\Pi^{\mathrm{st}}_d(0,z) \qquad (\mathrm{Re}\,z>0). \tag{S12.5}$$

For $2m_d \ge 4$, $\|R_d(s)\| \le C_d s^{2m_d-4}$ on $s>0$ forces $R_d(z) = z^{2m_d-4}G_{4,d}(z)$ with $G_{4,d}$ holomorphic near 0 (for $2m_d<4$, set $G_{4,d}(z) = z^{4-2m_d}R_d(z)$), proving $z^4\Pi^{\mathrm{st}}_d(0,z) = G_{4,d}(z) \to G_{4,d}(0)$ and $z^6\Pi^{\mathrm{st}}_d(0,z) = z^2G_{4,d}(z)\to 0$ across all branches.

Formal correspondence. Commits 71e, 9c73: `alpha-source/em-identification/ActualDressed{NoetherHalfIntegral,NonlinearPolarization,SignalPencil,FieldTimeFull,GradedSlopePrice,CubicNoetherPrice,CubicStaticPrice,IRAnalytic,CubicIRFactor,CubicIRReturn}.lean`.

S13 Theorem 7.5: Full-Field Quantum Feedback, Nine-Constraint Schur Equivalence, and Two-Clock Flux

With nine-slot injection $J_9 : \mathbb{C}^9 \to \mathbb{C}^{289}$ (slots 112–120, $r_9 J_9 = I_9$) and change-of-basis $F(P)$, set $N(P) = F(P)J_9$, $D(P) = r_9 F(P)^{-1}$, $C(P) = r_9 F(-P)^{\mathsf{T}} = N(-P)^{\mathsf{T}}$, and $L(P) = F(-P)^{-\mathsf{T}} J_9$, so $HN = 0, CH = 0, DN = CL = I_9$ and

$$GH = I - ND, \qquad HG = I - LC, \qquad DG = 0. \tag{S13.1}$$

1. **Half-axis nine-constraint Schur equivalence**: for $\det(I - GQ_d) \neq 0$, setting $A = (I - GQ_d)^{-1}$ and $x(c) = A(Gf + Nc)$ gives $x(c) = Nc + G(f + Q_d x(c))$ and $Dx(c) = c$. Left-multiplying by H via (S13.1) yields $[H - Q_d]x(c) = f - LC(f + Q_d x(c))$; since $CL = I_9$, vanishing is equivalent to $C(f + Q_d x(c)) = Sc - b = 0$. Conversely, $[H - Q_d]x = f$ gives $(I - GQ_d)x = Gf + NDx$, hence $x = x(Dx)$, and $C(f + Q_d x) = CHx = 0 \implies S(Dx) = b$ without assuming invertibility of S.
2. **Anchor bound** $16/15$ **and four-factor synchronous derivative**: choosing nonreal matter radius $r_d \geq 4(1 + G_*^{\text{price}} B_d^{\text{win}}) > 0$ gives $\|T^{-1} G_* \Pi_T\| \leq 1/16$ and $\|A_T\| \leq 16/15$ at $z = 3, 0 < T \leq 1$; differentiating $S_T(z) = C(z)\Pi_T(z)A_T(z)N(z)$ via $A'_T = -A_T B'_T A_T$ yields Eq. (7.8), while general two-clock half-axis fractions clear denominators by $p_d(\lambda)p_d(\lambda - c)$ and temporal Ward residuals vanish at $O(\epsilon)$.

Formal correspondence. Commit 9c73: `alpha-source/em-identification/{ActualEMDressedSourceAnchor,ActualEMDressedSchur,ActualDressedConstraintHistory,ActualDressedRationalFourier,ActualDressedTemporalQuantumResidual,ActualDressedFullSchurFlux,ActualDressedHalfSchur}.lean`.

S14 Theorem 7.11: Charged Colour-Singlet Spin-1/2 Three-Fermion State and Four-Channel Positive Spectrum

S14.1–S14.2 Exact Quantum Numbers and Bounded Four-Channel Sum

On each branch $d \in \{0, 1\}$, with $a^\dagger_{d,(s,c,h)}$ indexing spinor $s \in \{0, 1, 2, 3\}$, colour $c \in \{0, 1, 2\}$, and wedge type $h \in \{0, 1\}$, set

$$\epsilon_d(s_0, s_1, s_2) = \sum_{\pi \in S_3} \operatorname{sgn}(\pi)\, a^\dagger_{d,(s_0,\pi_0,0)} a^\dagger_{d,(s_1,\pi_1,0)} a^\dagger_{d,(s_2,\pi_2,1)} \Omega_d, \qquad q_d^{(3)} = \tfrac{1}{3}[\epsilon_d(0,1,2) - \epsilon_d(0,0,3)]. \tag{S14.1}$$

Orthogonality of the 6 unit words in $\epsilon_d(0, 1, 2)$ (squared norm 6) and 3 weight-2 words in $\epsilon_d(0, 0, 3)$ (squared norm 12) gives $\|q_d^{(3)}\|^2 = (6 + 12)/9 = 2$. Every $A \in \mathfrak{su}(3)$ acts on ϵ_{abc} by $\operatorname{tr} A = 0$ (colour singlet); electric charge is $(-1)^d = \pm 1$; Clifford spin-squared gives $\sum_{j=1}^3 (s_j^{(1)} + s_j^{(2)} + s_j^{(3)})^2 q_d^{(3)} = \frac{3}{4} q_d^{(3)}$ (spin 1/2); and $Y q_d^{(3)} \neq 0$.

Because $\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ preserve particle number and commute with v^{m+2}, $\mathcal{B} = \sum_{b=0}^{3} \mathcal{B}_b$ is bounded on $\mathcal{H}$, and the 57-term polynomial bound places $w_F(t) \in L^1 \cap L^2$, yielding Parseval's identity (7.19) $\nu(\mathbb{R}) = 2\pi \int_{\mathbb{R}} \|\mathcal{B} w_F(t)\|^2 dt$.

S14.3–S14.5 Imaginary Nonvanishing Witness, Meromorphic Real Window, Neutral Zero Channel, and Two-Resolvent Identity

At purely imaginary transfer $x = i$ ($k = p_L = p_R = 0$), the four channel amplitudes evaluate exactly to

$$\sqrt{30}\left(-\frac{1}{25}, \quad \frac{503044300641411654480741048304 33}{976991165662321560271023019875 00}, \quad -\frac{27}{316}, \quad 0\right), \tag{S14.2}$$

whose coherent sum is

$$\sqrt{30}\,\frac{150292586494789570208463523445404 1}{385911510436617016307054092850625 0} \neq 0. \tag{S14.3}$$

Because all 33 numerators and 17 denominators are meromorphic and nonzero at $x = i$, taking the minimum of their isolated Laurent zero/pole radii yields a **common punctured real interval** $0 < |x| < \delta$ **prior to all** $(F, \mu, \textbf{adjoint}, \textbf{causal})$, where $z\mathcal{B} r_F(z) \to -\mathcal{B}(q_d^{(3)} f) \neq 0$ forces strictly positive total spectral and causal time mass. Orthogonal grading $\int \|r_F(\omega)\|^2 d\omega = (\pi/\mu)\|q_d^{(3)} f\|^2 + \int \|\mathrm{corr}_{Y,F}(\omega)\|^2 d\omega$ with $B_Y(q_d^{(3)} f) \neq 0$ proves strictly positive integrated Yukawa excess, while $[Q_s, \mathcal{B}] = 0$ and $Q_s q_d^{(3)} = \pm 3 q_d^{(3)}$ annihilate $\langle e_w, \mathcal{B} r\rangle$ identically whenever $Q_s e_w = 0$.

Finally, at $\mu_* > 0$, the Sylvester solution $\mathcal{L}(D) = \sum_{a,b} [2\mu_* - i\,\mathrm{dir}(c)(e_a - e_b)]^{-1} P_a D P_b$ over $C_F = \sum_a e_a P_a$ yields via four-pole contour integration

$$\int_{\mathbb{R}} \|R_F(\omega) D R_F(\omega) g\|^2 d\omega = \frac{\pi}{\mu_*} \|\mathcal{L}(D) g\|^2 + \int_{\mathbb{R}} \|\mathcal{L}(D) R_F(\omega) g\|^2 d\omega, \tag{S14.4}$$

closing the signed frequency balance, pressure moment tail, and scalar field energy identity $(3c_\sharp/(16\mu_*) + 15c_\sharp/(16\mu_*) = 9c_\sharp/(8\mu_*))$.

Formal correspondence. Commits 71e, 9c73: `external-composite-decay/{MixedSpectatorYukawaNonzero,SourceYukawaPositiveCorrection}.lean`, `SourceActualFourBlock{Retarded,HilbertOperator,SharpCausalMass,RealTransferPositive,SignedSector}.lean`, and `Lean/scratch/LowEnergyRetardedCouncil/ActualFourBlockResponse/SourceActual{TwoResolventSylvester,ExactSylvesterRawFrequencyBalance,BalancedPressureMomentPayment,ScalarFieldDebitPayment}.lean`.

S15 Proposition 10.4: First-Registration Probability on the Complete 120 ns Gate and Primal/Adjoint Recovery

1. **(S15.1–S15.2) Trace-preserving free bath, first forcing, and factorial tail**: $\mathcal{L}_{\mathrm{nat}}(\rho) = K\rho + \rho K^\dagger + \sum_j J_j \rho J_j^\dagger$ (retaining same-arm and cross-arm recycling plus Hermitian cross-pairs) generates $\mathcal{U}_0(t,0)$ with Volterra expansion $\rho(t) = \mathcal{U}_0(t,0)\rho(0) + \int_0^t \mathcal{U}_0(t,s)\mathcal{R}(s)\rho(s) ds$ and simplex tail bound $\sum_{k\geq 0} a^{N+1+k}/(N+1+k)! \leq \frac{a^{N+1}}{(N+1)!(1-a/(N+2))}$ ($a < N+2$). The

ten-factor bath passes the 10^{-8} gate (joint uniform error 2.41×10^{-14}) over 40 local operators and 328 signed DAG rows.

2. **(S15.3) Four-port late adjoint**: $E_r^{\mathrm{late}} = \int_{I_{\mathrm{late}}} \mathcal{U}_0(t,0)^*(\mathcal{J}_r^* I)dt$ across 60 1 ns slices and $4,915,200$ Gaussian-integer degree-64 quadrature entries satisfies max operator error $8.427437 \times 10^{-12} < 10^{-10}$.
3. **(S15.4) Three no-count preimages, single initial-error contraction, and 10^{-21} primal/adjoint agreement**: from $1_{\mathrm{first}} = 1 - n_\perp - n_\parallel + n_\perp n_\parallel$ and $\mathcal{L}_Q^{\mathrm{nc}} = \mathcal{L}_{\mathrm{nat}} - \sum_{j\in Q} \mathcal{J}_j$, cyclic trace identities enforce $\frac{d}{dt}\operatorname{tr}(E_Q \rho_Q) = 0$. Source activity $\|E_{\mathrm{first}}\| \le \kappa \le 0.003740914455$ contracts initial error once: $|P_{\mathrm{first}} - \widehat{P}_{\mathrm{first}}| \le \kappa E_0 + E_{\mathrm{new}}$. Six adjoint flows certify $[3.17845500810, 3.17846466595] \times 10^{-6}$ (error 4.82892×10^{-12}) and 30 primal curves (3600 slices) certify $[3.17845717771, 3.17846249634] \times 10^{-6}$ (error 2.65931×10^{-12}), with centre difference 1.301828×10^{-21}.

S15.5 Frozen Scientific Receipts and Commits

Module	Frozen scientific commit	Verified scope and precision gate
Four-port late adjoint `rha0038.1`	`a165b04a26febd0bc4e7ed1f33d4668b564a6e4f`	60 slices / $4,915,200$ Gaussian integers; passes 10^{-10} gate
Three no-count adjoint flows `rha0039`	`425ed6ed406b2a96f044dafd1c39b3587a9f14f0`	Six complete 120 ns adjoint flows; passes 10^{-8} gate
Single whole-event contraction `rha0039.1`	`87f9b1234441783f945727408b06ede72af9683d`	$\kappa \le 0.0037409145$ and adjoint probability interval
30 primal curves & recovery `rha0041.1`	Sci. `065399a21dcf...`; cons. `f7a5ee523ac8...`	3600 slices / 90 rational slices; primal interval & 1.30×10^{-21} centre match
Seven-query program bank `rha0042`	`7c9423057a2ee8f594b7502172e1a37105490829`	All 7 priority queries / 140 activation & scalar synthesis programs

Formal correspondence. `munich/readout-domain/` (`RegisteredDuhamelTail.lean`, `RegisteredTensorAction.lean`, `LateRegisteredAdjoint.lean`, `FirstReceiptZeroCount.lean`, `FirstReceiptQuantumFlux.lean`, `NoCountQuantumGenerator.lean`) and independent checkers.

S16 Theorems 4.6–4.7: Source-Generated Volterra Uniqueness and Derivative-Loss-Free Symmetric-Hyperbolic Closure

S16.1 Bidirectional Equivalence of Weighted Mother-Action Weak Zeros and Volterra Fixed Points

Fix $T > 0$, spatial box $Q = [a,b] \subset \mathbb{R}^3$ $(a \le b)$, and dense spatial test element ϕ_n $(n \in \mathbb{N})$. Continuity of the source-generated whole-time L^2 trajectory and bounded operator pairing render $m_n(t) = \langle \phi_n, M(t) u_{\mathrm{src}}(t) \rangle_{L^2(Q)}$ and the spatial Green rate $g_n(t)$ interval-integrable on $[0,T]$. Setting $G_n(t) = m_{n,0} + \int_0^t g_n(s)\,ds$, G_n is absolutely continuous on $[0,T]$ with $G_n'(t) = g_n(t)$ for a.e. $t \in [0,T]$.

For any smooth weight $w \in C^1([0,T])$ satisfying $w(T) = 0$, integrating the absolutely continuous product $w(t)G_n(t)$ by parts gives

$$\int_0^T (w(t)g_n(t) + w'(t)G_n(t))\,dt = \big[w(t)G_n(t)\big]_0^T = 0 - w(0)m_{n,0} = -w(0)m_{n,0}. \tag{S16.1}$$

- **Volterra fixed point $\Longrightarrow$ weighted weak-action zero**: if $m_n(t) = G_n(t)$ a.e. on $[0,T]$, then $\int_0^T w'(t)m_n(t)\,dt = \int_0^T w'(t)G_n(t)\,dt$, and (S16.1) yields $\int_0^T (wg_n + w'm_n)\,dt + w(0)m_{n,0} = 0$ for every $w \in C^1([0,T])$ with $w(T) = 0$.
- **Weighted weak-action zero $\Longrightarrow$ Volterra fixed point**: conversely, subtracting (S16.1) from the weak-action zero identity gives for $\Delta_n(t) = m_n(t) - G_n(t) \in L^1([0,T])$

$$\forall w \in C^1([0,T])\ (w(T) = 0), \qquad \int_0^T w'(t)\,\Delta_n(t)\,dt = 0. \tag{S16.2}$$

  For any $\varphi \in C_c^\infty((0,T))$, the right-anchored primitive $w_\varphi(t) = -\int_t^T \varphi(s)\,ds$ belongs to $C^1([0,T])$, satisfies $w_\varphi(T) = 0$, and has $w_\varphi'(t) = \varphi(t)$ (with $w_\varphi(0)$ unconstrained). Hence $\int_0^T \varphi(t)\Delta_n(t)\,dt = 0$ for all $\varphi \in C_c^\infty((0,T))$, which by the fundamental lemma of the calculus of variations forces $\Delta_n(t) = 0$ a.e. on $[0,T]$, proving Theorem 4.6(1).

S16.2 Common Full-Measure Time Set Across Countable Dense Tests and Spatial $L^2(Q)$ Uniqueness

For each $n \in \mathbb{N}$, define

$$\Omega_n = \{t \in [0,T] :\ m_n(t) = G_n(t)\}, \qquad \Omega_{T,Q} = \bigcap_{n\in\mathbb{N}} \Omega_n \subseteq [0,T]. \tag{S16.3}$$

By S16.1, each complement $[0,T] \setminus \Omega_n$ has Lebesgue measure zero; countable unions of null sets are null, so $\Omega_{T,Q}$ has full measure $T > 0$ on $[0,T]$ and is therefore nonempty.

Fix any $t \in \Omega_{T,Q}$ and candidate $v \in L^2(Q)$. If $\langle \phi_n, M(t)v\rangle_{L^2(Q)} = G_n(t)$ for every $n \in \mathbb{N}$, then $t \in \Omega_n$ gives $G_n(t) = m_n(t) = \langle \phi_n, M(t)u_{\mathrm{src}}(t)\rangle_{L^2(Q)}$, so $w = v - u_{\mathrm{src}}(t) \in L^2(Q)$ satisfies $\langle \phi_n, M(t)w\rangle_{L^2(Q)} = 0$ for all $n \in \mathbb{N}$. Density of $\mathrm{span}\{\phi_n\}$ implies $M(t)w = 0$, and positive-definite invertibility of $M(t)$ on the Cauchy-safe background forces $w = 0$, i.e. $v = u_{\mathrm{src}}(t)$, proving Theorem 4.6(2).

S16.3 Recursive Spatial-Jet Prolongation and On-Shell-Clean Commuted Forcing

Write $\mathcal{L}_{\mathrm{mat}}u = A^0(x)\partial_t u + \sum_{j=1}^3 A^j(x)\partial_j u + B_0(x)u$ with smooth invertible $A^0(x)$, so any on-shell field satisfies

$$\mathcal{L}_{\mathrm{mat}}u = 0 \quad \Longrightarrow \quad \partial_t u = \mathcal{T}(x)(j^1_{\mathrm{sp}}u) \equiv -(A^0(x))^{-1}\Big(\sum_{j=1}^3 A^j(x)\partial_j u + B_0(x)u\Big). \tag{S16.4}$$

Inducting on $|w|$ for $w \in \{1,2,3\}^*$, we construct $\mathcal{J}_w(x) \in \mathcal{L}(J_{\mathrm{sp}}^{|w|}, \mathcal{V})$ purely from source coefficients:

1. **Base case** $w = \emptyset$: set $\mathcal{J}_\emptyset(x) \equiv 0$, so $[\partial^\emptyset, \mathcal{L}_{\mathrm{mat}}]u = 0 = \mathcal{J}_\emptyset(x)(j^0_{\mathrm{sp}}u)$.
2. **Inductive step** $w = i\cdot\tilde{w}$ ($|\tilde{w}| = k$, $|w| = k+1$): for any smooth solution of $\mathcal{L}_{\mathrm{mat}}u = 0$, expand

$$[\partial_i\partial^{\tilde{w}}, \mathcal{L}_{\mathrm{mat}}]u = \partial_i([\partial^{\tilde{w}}, \mathcal{L}_{\mathrm{mat}}]u) + [\partial_i, \mathcal{L}_{\mathrm{mat}}](\partial^{\tilde{w}}u). \tag{S16.5}$$

Differentiating the inductive identity $[\partial^{\tilde{w}}, \mathcal{L}_{\mathrm{mat}}]u = \mathcal{J}_{\tilde{w}}(x)(j^k_{\mathrm{sp}}u)$ along x^i gives

$$\partial_i(\mathcal{J}_{\tilde{w}}(x)(j^k_{\mathrm{sp}}u)) = (\partial_i\mathcal{J}_{\tilde{w}})(x)(j^k_{\mathrm{sp}}u) + \mathcal{J}_{\tilde{w}}(x)(\mathsf{S}_i^{(k)}(j^{k+1}_{\mathrm{sp}}u)), \tag{S16.6}$$

where $\mathsf{S}_i^{(k)}$ reads $\partial^{i\cdot v}u$ ($|v| \le k$) from $j^{k+1}_{\mathrm{sp}}u$. Meanwhile, the second term of (S16.5) reads

$$[\partial_i, \mathcal{L}_{\mathrm{mat}}](\partial^{\tilde{w}}u) = (\partial_iA^0)(x)\,\partial_t\partial^{\tilde{w}}u + \sum_{j=1}^{3}(\partial_iA^j)(x)\,\partial_j\partial^{\tilde{w}}u + (\partial_iB_0)(x)\,\partial^{\tilde{w}}u. \tag{S16.7}$$

Applying $\partial^{\tilde{w}}$ to $\mathcal{L}_{\mathrm{mat}}u = 0$ yields $\mathcal{L}_{\mathrm{mat}}(\partial^{\tilde{w}}u) = -[\partial^{\tilde{w}}, \mathcal{L}_{\mathrm{mat}}]u = -\mathcal{J}_{\tilde{w}}(x)(j^k_{\mathrm{sp}}u)$, solving for the mixed temporal–spatial derivative strictly in terms of $(k+1)$-st pure spatial jets:

$$\partial_t\partial^{\tilde{w}}u = -(A^0(x))^{-1}\Big(\sum_{j=1}^{3}A^j(x)\,\partial_{j\cdot\tilde{w}}u + B_0(x)\,\partial^{\tilde{w}}u + \mathcal{J}_{\tilde{w}}(x)(j^k_{\mathrm{sp}}u)\Big). \tag{S16.8}$$

Substituting (S16.8) into (S16.7) and adding (S16.6) defines purely from source coefficients

$$\begin{aligned}\mathcal{J}_{i\cdot\tilde{w}}(x)(J) = {}&(\partial_i\mathcal{J}_{\tilde{w}})(x)(\pi_kJ) + \mathcal{J}_{\tilde{w}}(x)(\mathsf{S}_i^{(k)}J) + \sum_{j=1}^{3}(\partial_iA^j)(x)J_{j\cdot\tilde{w}} + (\partial_iB_0)(x)J_{\tilde{w}}\\ &- (\partial_iA^0)(x)(A^0(x))^{-1}\Big(\sum_{j=1}^{3}A^j(x)J_{j\cdot\tilde{w}} + B_0(x)J_{\tilde{w}} + \mathcal{J}_{\tilde{w}}(x)(\pi_kJ)\Big),\end{aligned} \tag{S16.9}$$

which belongs to $\mathcal{L}(J^{k+1}_{\mathrm{sp}}, \mathcal{V})$ and satisfies $\mathcal{J}_w(x)(j^{|w|}_{\mathrm{sp}}u(x)) = [\partial^w, \mathcal{L}_{\mathrm{mat}}]u(x)$ pointwise for all smooth on-shell u, proving Theorem 4.7(1).

S16.4 Compact-Box Coercivity, Wordwise Square Absorption, and Derivative-Loss-Free Rate Closure

Fix $I \times Q = [t_0, t_1] \times [a, b]$ and $r \in \mathbb{N}$.

1. **Pointwise coercivity**: continuity and fibrewise positive-definiteness of $A^0(t,\mathbf{x})$ on the compact product $(I \times Q) \times S(\mathcal{V})$ give $\kappa > 0$ such that $\kappa\|\xi\|^2 \le \langle\xi, A^0(t,\mathbf{x})\xi\rangle$. Under the sup-word jet norm $\|j^r_{\mathrm{sp}}u(x)\| = \max_{|w|\le r}\|\partial^wu(x)\|$,

$$\kappa\,\|j^r_{\mathrm{sp}}u(t,\mathbf{x})\|^2 \;\le\; \sum_{|w|\le r}\langle\partial^wu(t,\mathbf{x}), A^0(t,\mathbf{x})\partial^wu(t,\mathbf{x})\rangle = \mathcal{E}_r[u](t,\mathbf{x}). \tag{S16.10}$$

2. **Budget absorption prior to all fields**: set $M_{\mathrm{inv}} = \sup_{I\times Q}\|(A^0)^{-1}\| < \infty$, $M_{\mathrm{jet}} = \max_{|w|\le r}\sup_{I\times Q}\|\mathcal{J}_w\| < \infty$, $N_r = \sum_{k=0}^{r}3^k$, and $L = N_r(M_{\mathrm{inv}}M_{\mathrm{jet}})^2 \ge 0$ (prior to all u).

For any smooth on-shell u and $|w| \le r$, monotonicity $\|j_{\mathrm{sp}}^{|w|}u\| \le \|j_{\mathrm{sp}}^{r}u\|$ and Theorem 4.7(1) yield

$$\left\|(A^0)^{-1}[\partial^w, \mathcal{L}_{\mathrm{mat}}]u\right\| = \left\|(A^0)^{-1}\mathcal{J}_w(j_{\mathrm{sp}}^{|w|}u)\right\| \le M_{\mathrm{inv}}M_{\mathrm{jet}}\|j_{\mathrm{sp}}^{r}u\|. \tag{S16.11}$$

Squaring, summing over all N_r words, multiplying by κ, and applying (S16.10) gives

$$\kappa\,\mathcal{F}_r[u](t,\mathbf{x}) \le \kappa\,N_r(M_{\mathrm{inv}}M_{\mathrm{jet}})^2\|j_{\mathrm{sp}}^{r}u(t,\mathbf{x})\|^2 \le L\,\mathcal{E}_r[u](t,\mathbf{x}). \tag{S16.12}$$

3. **Same-order energy-rate closure**: applying the symmetric-hyperbolic pointwise identity to $\partial^w u$ bounds $(\partial_t A^0)$, $\sum_j(\partial_j A^j)$, zero-order symmetric terms $-(B_0 + B_0^{\mathsf{T}})$, and $2\langle\partial^w u, [\partial^w, \mathcal{L}_{\mathrm{mat}}]u\rangle$ linearly in $\|j_{\mathrm{sp}}^{r}u\|^2$ and $\mathcal{F}_r[u]$; combining compact-box coefficient bounds with (S16.10) and (S16.12) into a single $K \ge 0$ yields $\kappa\,\mathcal{R}_r[u](t,\mathbf{x}) \le K\,\mathcal{E}_r[u](t,\mathbf{x})$ with zero $(r+1)$-st derivatives, proving Theorem 4.7(2).

Formal correspondence. Commit H: `Lean/SaturationMonoid/PhysicsCore/StageNineDiracDualFormNativeFixedP506CauchySafeMatterCanonicalAffineVolterra{Recognition,Actualization}.lean` (`weakVolterraActionZero_iff_fixedPoint`, `canonicalAffinePhysicalTimeL2Output_weakActionZero_iff_fixedPoint`, `CanonicalAffineVolterraDenseReadOccurrence.physicalField_unique`) and `StageNineDiracDualFormNativeFixedP506CauchySafeMatterSymmetricHyperbolicAllOrder{JetProlongation,EnergyClosure}.lean` (`fixedMatterAllOrderForcingJetCLM_apply_of_actionZero`, `exists_fixedMatterAllOrderPointwiseForcingBudgetAbsorptionOnBox`, `exists_fixedMatterAllOrderClosedPointwiseRateBoundOnBox`).

S17 Theorems 7.8 and 7.12: 192/97 Electromagnetic Fraction, Joint Cone, and 4D Curvature Strong Integration

S17.1 192-Dimensional Inactive Two-Sided Inverse and 97-Dimensional Bivariate Schur Fraction

For dressed event d, unit n ($|n|^2 = 1$), $0 < \varepsilon \le 1/2$, and $\sigma > 0$, split $289 = 192 + 97$ into inactive and active directions:

$$P_{\mathrm{corr}}(\varepsilon,\sigma) = \begin{pmatrix} P_{II}(\varepsilon,\sigma) & P_{IA}(\varepsilon,\sigma) \\ P_{AI}(\varepsilon,\sigma) & P_{AA}(\varepsilon,\sigma) \end{pmatrix}, \qquad P_{II} \in M_{192}(\mathbb{C}), \quad P_{AA} \in M_{97}(\mathbb{C}). \tag{S17.1}$$

On $0 < \varepsilon \le 1/2, \sigma > 0$, $P_{II}^{-1} = d_{\mathrm{nat}}(\varepsilon,\sigma)^{-1}N_{\mathrm{inv}}(\varepsilon,\sigma)$ with $d_{\mathrm{nat}} \ne 0$. Block Gauss elimination gives $\det P_{\mathrm{corr}} = \det P_{II}\cdot\det S_{97}$ for $S_{97} = P_{AA} - P_{AI}P_{II}^{-1}P_{IA}$ (retaining all $97 \leftrightarrow 192$ cross-columns) and reconstructs $x_I = -P_{II}^{-1}(P_{IA}x_A - f_I)$, making $\det_{\mathrm{corr}} \ne 0 \iff \det S_{97} \ne 0$.

Clearing d_{nat} defines the bivariate polynomial matrix

$$Q_{97}(\varepsilon,X) = d_{\mathrm{nat}}(\varepsilon,X)P_{AA}(\varepsilon,X) - P_{AI}(\varepsilon,X)N_{\mathrm{inv}}(\varepsilon,X)P_{IA}(\varepsilon,X) \in M_{97}((\mathbb{C}[\varepsilon])[X]), \tag{S17.2}$$

whose adjugate $Q_{97}\,\mathrm{adj}(Q_{97}) = (\det Q_{97})I_{97}$ expresses $\varepsilon^4 T_{00}(\varepsilon,\sigma) = \mathrm{eval}_{\varepsilon,\sigma}(N_{97})/\mathrm{eval}_{\varepsilon,\sigma}(\det Q_{97})$, proving Theorem 7.8(1).

S17.2 Elementary Row Update at Electromagnetic Pivot 15 and X^2/X^3 Clock Cancellation

The physical electromagnetic temporal vector vanishes on all 192 inactive slots ($E_I = 0$) and on the 97-element active index set $\mathcal{J}_{97} \subset \{0,\dots,288\}$ takes exact rational value $(E_A)_{15} = -1/2$ at active pivot 15. Temporal Ward cancellation annihilates the X^0, X^1 terms of both $E_A^{\mathsf{T}} P_{AA}$ and $E_A^{\mathsf{T}} P_{AI} N_{\mathrm{inv}} P_{IA}$, giving

$$E_A^{\mathsf{T}}\, Q_{97} = X^2\, R_{97}^{\mathsf{T}}, \qquad R_{97} \in \big((\mathbb{C}[\varepsilon])[X]\big)^{97}. \tag{S17.3}$$

Replacing the pivot-15 row of Q_{97} by $E_A^{\mathsf{T}} Q_{97} = \sum_{j\in\mathcal{J}_{97}} (E_A)_j (Q_{97})_{j,\bullet}$ to form $\widetilde{Q}_{97}$ gives by row multilinearity and alternating cancellation

$$-\frac{1}{2}\det Q_{97} = (E_A)_{15}\det Q_{97} = \det \widetilde{Q}_{97} = \det(\text{row 15 replaced by } X^2 R_{97}^{\mathsf{T}}) = X^2\,\det \widehat{Q}_{97}. \tag{S17.4}$$

Multiplying by -2 with $D_{\mathrm{W}} = -2\det \widehat{Q}_{97} \in (\mathbb{C}[\varepsilon])[X]$ proves $\det Q_{97} = X^2 D_{\mathrm{W}}$ inside the polynomial ring without division, while the auxiliary clock column yields $N_{97} = X^3 N_{\mathrm{W}}$. For $\sigma > 0$, ring evaluation gives

$$\frac{\mathrm{eval}_{\varepsilon,\sigma}(N_{97})}{\mathrm{eval}_{\varepsilon,\sigma}(\det Q_{97})} = \frac{\sigma^3\,\mathrm{eval}_{\varepsilon,\sigma}(N_{\mathrm{W}})}{\sigma^2\,\mathrm{eval}_{\varepsilon,\sigma}(D_{\mathrm{W}})} = \sigma\,\frac{\mathrm{eval}_{\varepsilon,\sigma}(N_{\mathrm{W}})}{\mathrm{eval}_{\varepsilon,\sigma}(D_{\mathrm{W}})}, \tag{S17.5}$$

and domain integrity of $(\mathbb{C}[\varepsilon])[X]$ yields $\mathrm{ord}_X(P) = m + \mathrm{ord}_X(W)$ for $m = 3, 2$, proving Theorem 7.8(2).

S17.3 Bivariate Leading-Coefficient Nonvanishing and Single-Seed Joint Cone

Given one seed $(\varepsilon_0, \sigma_0)$ with $0 < \varepsilon_0 \le 1/2$, $\sigma_0 > 0$, and $\det_{\mathrm{corr}}(\varepsilon_0,\sigma_0) \neq 0$, the denominator $B = \det Q_{97} \in (\mathbb{C}[\varepsilon])[X]$ is nonzero; write $A = N_{97} \in (\mathbb{C}[\varepsilon])[X]$. For each $P \in \{A, B\}$, factoring the X-trailing order $k_P = \mathrm{ord}_X(P)$ (with convention $k_P = 0, p_0 \equiv 0$ when $P = 0$) writes

$$P(\varepsilon, X) = X^{k_P}\Big(p_0(\varepsilon) + X\,P_{\mathrm{tail}}(\varepsilon, X)\Big). \tag{S17.6}$$

Factoring $\ell_P = \mathrm{ord}_\varepsilon(p_0)$ writes $p_0(\varepsilon) = \varepsilon^{\ell_P}(c_P + \varepsilon\tilde{p}(\varepsilon))$ (setting $k_A = \ell_A = 0, a_0 \equiv 0, c_A = 0$ if $A = 0$). Since $B \neq 0$, $b_0(\varepsilon) \neq 0$ and $c_B \neq 0$. Define the spatial rescaling weight $w_{\mathrm{sp}}(\varepsilon) = \varepsilon^{\ell_B - \ell_A}$ and, denoting by $M_{\tilde{a}}, M_{\tilde{b}} \ge 0$ the sums of coefficient moduli of $\tilde{a}, \tilde{b}$ on $[0, 1/2]$, set $S_{\mathrm{ratio}} := 2|c_B|^{-1}(M_{\tilde{a}} + |c_A||c_B|^{-1} M_{\tilde{b}}) \ge 0$. Finite polynomial coefficient bounds select $\rho_{\mathrm{sp}} \in (0, 1/2]$ so that $|\varepsilon\tilde{b}(\varepsilon)| \le \frac{1}{2}|c_B|$ for all $0 < \varepsilon \le \rho_{\mathrm{sp}}$, and then for each fixed $\varepsilon \in (0, \rho_{\mathrm{sp}}]$ select $\rho_{\mathrm{clk}}(\varepsilon) > 0$ so that for all $0 < \sigma \le \rho_{\mathrm{clk}}(\varepsilon)$: (1) $|\sigma B_{\mathrm{tail}}(\varepsilon,\sigma)| \le \frac{1}{2}|b_0(\varepsilon)|$, certifying $\det_{\mathrm{corr}}(\varepsilon,\sigma) \neq 0$; and (2) $|\widetilde{R}(\varepsilon,\sigma) - a_0(\varepsilon)/b_0(\varepsilon)| \cdot |w_{\mathrm{sp}}(\varepsilon)| \le \varepsilon$ for $\widetilde{R}(\varepsilon,\sigma) = \sigma^{k_B - k_A}\varepsilon^4 T_{00}(\varepsilon,\sigma)$. Combining this with $|w_{\mathrm{sp}}(\varepsilon)a_0(\varepsilon)/b_0(\varepsilon) - c_A/c_B| = \varepsilon|(\tilde{a}c_B - c_A\tilde{b})/[c_B(c_B + \varepsilon\tilde{b})]| \le S_{\mathrm{ratio}}\,\varepsilon$ via the triangle inequality yields (7.14), proving Theorem 7.8(3).

S17.4 Four-Dimensional Curvature Strong Integration and $\mathrm{Dom}(H_{\mathrm{time}})$ Word Gram

Let $\mathcal{U}_{1\text{–}10}$ be the ten-stage total closure object with Standard Model connection A_{cl}, let $\eta \in C_c^\infty(\mathbb{R}^4, \Omega^1(\mathfrak{h}_{\mathrm{SM}}))$ and $t \in \mathbb{R}$, and denote by H_{time} the self-adjoint generator of the strongly continuous unitary clock group $U_{\mathrm{time}}(s) = e^{-isH_{\mathrm{time}}}$ on the 252-dimensional matter response space and common carrier $\mathcal{H}_{\mathrm{com}}$ (distinct from the nonsymmetric 504-mode CAR generator $H = H_0 + Y$ of §6.3). For $A_{\mathrm{cl}} + t\eta$, $F(A_{\mathrm{cl}} + t\eta) = F(A_{\mathrm{cl}}) + t\, D_{A_{\mathrm{cl}}}\eta + \frac{t^2}{2}[\eta \wedge \eta]$ defines on $\pi_{\mathrm{sp}}(p) = (0, \mathbf{p})$ a uniformly bounded strongly continuous seed, making the 4D Bochner integral (7.20) well-defined for $f \in \mathcal{S}(\mathbb{R}^4) \subset L^1(\mathbb{R}^4)$ with $\|\Phi_{\eta,t,P}(f)\| \le \|f\|_{L^1}\|\mathrm{seed}\|_\infty$ and, at $t = 0$, $\Phi_{\eta,0,P}(f) = \Phi_{\mathrm{orig}}(\iota_0(P), f)$ for the origin embedding $\iota_0(P)$ of the same-point curvature polynomial P.

Using $U_{\mathrm{time}}(s)U_{\mathrm{time}}(-p_0) = U_{\mathrm{time}}(-(p_0 - s))$ and $\pi_{\mathrm{sp}}(p_0 - s, \mathbf{p}) = \pi_{\mathrm{sp}}(p)$, substituting $q_0 = p_0 - s$ gives the covariance relation with $\tau_s f(q_0, \mathbf{p}) = f(q_0 + s, \mathbf{p})$

$$U_{\mathrm{time}}(s)\, \Phi_{\eta,t,P}(f)\, U_{\mathrm{time}}(-s)v = \Phi_{\eta,t,P}(\tau_s f)v. \tag{S17.7}$$

Since $s^{-1}(\tau_s f - f) \to \partial_0 f$ in $L^1(\mathbb{R}^4)$, $\frac{d}{ds}\big|_{s=0}(U_{\mathrm{time}}(s)\Phi(f)U_{\mathrm{time}}(-s)) = \Phi(\partial_0 f)$ in operator norm. For $v \in \mathrm{Dom}(H_{\mathrm{time}})$,

$$\frac{U_{\mathrm{time}}(s)\Phi(f)v - \Phi(f)v}{s} = \frac{(U_{\mathrm{time}}(s)\Phi(f)U_{\mathrm{time}}(-s) - \Phi(f))}{s}U_{\mathrm{time}}(s)v + \Phi(f)\frac{U_{\mathrm{time}}(s)v - v}{s} \tag{S17.8}$$

converges strongly as $s \to 0$ to $\Phi(\partial_0 f)v - i\Phi(f)H_{\mathrm{time}}v$, proving $\Phi_{\eta,t,P}(f)v \in \mathrm{Dom}(H_{\mathrm{time}})$ with $H_{\mathrm{time}}\Phi_{\eta,t,P}(f)v = \Phi_{\eta,t,P}(i\partial_0 f)v + \Phi_{\eta,t,P}(f)H_{\mathrm{time}}v$ (and inductively for all finite mixed words $W(w)$). Pulling back via $I_{\mathrm{com}} : \mathcal{H}_{\mathrm{com}} \xrightarrow{\sim} \mathcal{H}_{\mathrm{src}}$ and applying Theorem 6.2 yields $\sum_{i,j}\overline{c_i}c_j\mathrm{Gram}_x(w_i, w_j) = \left\|\sum_i c_i W(w_i)\psi_{\mathrm{prep}}(x)\right\|^2 \ge 0$, proving Theorem 7.12.

Formal correspondence. Commit df4: `Verification/physics/low-energy-phenomenology/alpha-source/em-identification/ActualSourcePrepared{InactiveSchurReturn,Schur97NoetherFraction,Schur97WardSchur,Schur97WardDeterminant,Schur97WardFactor,Schur97JointCone}.lean` (`actual_source_prepared_em_enclosure_certification.json`). Commit 7e4: `Lean/SaturationMonoid/PhysicsCore/YangMills/SourceQuantum/Curvature/Spatial/{Seed,Field,Words,Original}.lean`, `Continuum/Spacetime/Strong/Spatial/{Field,Time}.lean` (`source_spatial_curvature_receipt.json`).

S18 Theorem 10.1: Whole-Word Kraus–Born Isometry and Open Recycling Density Trace Duality

S18.1 Exact Total Mass and L^2 Field-State Isometric Embedding on Simplex Direct Sums

On $[t_L, t_R]$ ($t_0 \in [t_L, t_R]$) with finite channel set $\mathcal{J}$, for $w = (j_1, \dots, j_n) \in \mathcal{J}^n$ and $\Delta_n(t_0, t_R) = \{t_0 \le s_1 \le \cdots \le s_n \le t_R\}$, the source Kraus word amplitude is

$$\mathcal{A}_{n,w}(t_0; \mathbf{s}) = U_{\mathrm{eff}}(t_R, s_n)J_{j_n}U_{\mathrm{eff}}(s_n, s_{n-1})\cdots J_{j_1}U_{\mathrm{eff}}(s_1, t_0) \in \mathcal{L}(E), \tag{S18.1}$$

where $\frac{d}{dt}U_{\text{eff}}(t,s) = K(t)U_{\text{eff}}(t,s)$ and $K(t)^* + K(t) + \sum_{j\in\mathcal{J}} J_j(t)^* J_j(t) = 0$. Integration by parts across simplices yields

$$\sum_{n=0}^{\infty} \sum_{w\in\mathcal{J}^n} \int_{\Delta_n(t_0,t_R)} \left\| \mathcal{A}_{n,w}(t_0;\mathbf{s})x \right\|^2 d\mathbf{s} = \|x\|^2. \tag{S18.2}$$

Piecewise continuity on compact Δ_n makes $\Psi_{t_0}(x)$ strongly measurable on $\mathcal{W}(\mathcal{J}) = \bigsqcup_{n,w} \Delta_n$, and Tonelli's theorem expands $\int_{\mathcal{W}} \|\Psi_{t_0}(x)\|^2 d\mu_{\text{base}}$ into (S18.2) to equal $\|x\|^2 < \infty$, proving $\|\Psi_{t_0}(x)\|_{L^2} = \|x\|$ and Theorem 10.1(1).

S18.2 Vanishing Derivative Against the Backward Adjoint Flow and All-Order Open Density Trace Identity

For any $\mathcal{H} \subseteq \mathcal{J}$, consider on the operator algebra $\mathcal{L}(E)$ of the finite-dimensional atomic Hilbert space E the forward open density IVP and backward Heisenberg effect terminal problem ($t \in [t_0, t_R]$):

$$\begin{cases} \dfrac{d}{dt}\Phi^{\mathcal{H}}_{t_0\to t}(\rho) = K(t)\Phi^{\mathcal{H}}_{t_0\to t}(\rho) + \Phi^{\mathcal{H}}_{t_0\to t}(\rho)K(t)^* + \displaystyle\sum_{j\in\mathcal{H}} J_j(t)\Phi^{\mathcal{H}}_{t_0\to t}(\rho)J_j(t)^*, \quad \Phi^{\mathcal{H}}_{t_0\to t_0}(\rho) = \rho, \\ -\dfrac{d}{dt}\mathcal{C}^{\mathcal{H}}_{t\to t_R}(A) = K(t)^*\mathcal{C}^{\mathcal{H}}_{t\to t_R}(A) + \mathcal{C}^{\mathcal{H}}_{t\to t_R}(A)K(t) + \displaystyle\sum_{j\in\mathcal{H}} J_j(t)^*\mathcal{C}^{\mathcal{H}}_{t\to t_R}(A)J_j(t), \quad \mathcal{C}^{\mathcal{H}}_{t_R\to t_R}(A) = A. \end{cases} \tag{S18.3}$$

1. **Bilinear trace conservation and whole-word expansion**: for arbitrary $\rho, A \in \mathcal{L}(E)$, differentiating $F(t) = \operatorname{Tr}\big(\Phi^{\mathcal{H}}_{t_0\to t}(\rho)\,\mathcal{C}^{\mathcal{H}}_{t\to t_R}(A)\big)$ cancels term by term via trace cyclicity:

$$F'(t) = \operatorname{Tr}\left(\frac{d\Phi}{dt}\,\mathcal{C} + \Phi\,\frac{d\mathcal{C}}{dt}\right) = 0 \qquad \forall t \in [t_0, t_R], \tag{S18.4}$$

   equating $\operatorname{Tr}(\Phi^{\mathcal{H}}_{t_0\to t_R}(\rho)A) = F(t_R) = F(t_0) = \operatorname{Tr}(\rho\,\mathcal{C}^{\mathcal{H}}_{t_0\to t_R}(A))$. Iterating Duhamel along U_{eff} expands $\mathcal{C}^{\mathcal{H}}_{t_0\to t_R}(A) = \sum_{n=0}^{\infty} \mathcal{E}^{\mathcal{H}}_n(t_0;A) = \mathcal{E}^{\mathcal{H}}_{\text{end}}(t_0;A)$ in operator norm, proving (10.2).
2. **Positivity, trace non-increase, and whole-bath trace preservation**: decomposing $\rho = \sum_\alpha |x_\alpha\rangle\langle x_\alpha| \geq 0$ gives $\operatorname{Tr}(\Phi^{\mathcal{H}}_{t_0\to t_R}(\rho)A) \geq 0$ for all $A \geq 0$, hence $\Phi^{\mathcal{H}}_{t_0\to t_R}(\rho) \geq 0$; setting $A = I$ with $\mathcal{H} \subseteq \mathcal{J}$ yields via (S18.2)

$$\begin{aligned} \operatorname{Tr}\Phi^{\mathcal{H}}_{t_0\to t_R}(\rho) &= \sum_\alpha \sum_{n=0}^{\infty} \sum_{w\in\mathcal{H}^n} \int_{\Delta_n} \|\mathcal{A}_{n,w}x_\alpha\|^2 d\mathbf{s} \\ &\leq \sum_\alpha \sum_{n=0}^{\infty} \sum_{w\in\mathcal{J}^n} \int_{\Delta_n} \|\mathcal{A}_{n,w}x_\alpha\|^2 d\mathbf{s} = \sum_\alpha \|x_\alpha\|^2 = \operatorname{Tr}\rho; \end{aligned}$$

   and at $\mathcal{H} = \mathcal{J}$, $\mathcal{C}^{\mathcal{J}}_{t\to t_R}(I) \equiv I$ (since $K^* + K + \sum_{j\in\mathcal{J}} J_j^* J_j = 0$) gives $\operatorname{Tr}\Phi^{\mathcal{J}}_{t_0\to t_R}(\rho) = \operatorname{Tr}\rho$ for all $\rho \in \mathcal{L}(E)$, proving Theorem 10.1(2).

Formal correspondence. Commits B291 and B300 respectively bind `Verification/physics/stage10/independent-bell/munich/readout-domain/evidence/source-kraus-word-wave-rha0291/source-r0004/SourceKrausWordWave291.lean` (`wordFunction_measurable`, `lintegral_bornDensity`,

bornMeasure_univ, sourceIsometry) and source-endpoint-recycling-rha0300/density-source-r0001/SourceEndpointDensity300.lean (whole_word_trace_hasSum, whole_word_trace_eq_recycling_IVP, densityFlow_positive, densityFlow_trace_nonincreasing, whole_bath_trace_identity).